



# Investigating the impacts of thermochemical group additivity values on kinetic model predictions through sensitivity and uncertainty analyses

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## ABSTRACT

The accuracy of species thermochemical properties is important for predicting combustion characteristics by chemical kinetic models due to their impacts on chemical equilibria of elementary reactions and on backward rate coefficients. As they are frequently estimated based on the group additivity method, the underlying group values directly influence the model prediction accuracy. While a number of studies have recently aimed to improve the accuracy of group values, the dependence of kinetic model predictions on the different groups has not yet been investigated in detail before. In the present work, the impact of group thermochemical parameters on ignition prediction is systematically analyzed by means of sensitivity analysis, employing the oxidation of 2-methylhexane (2-MH) as an example. Very high sensitivities are observed at intermediate temperatures for the peroxy group OO/C/H and its adjacent groups in low-temperature intermediate species. Based on uncertainty analysis, optimization potentials are then determined for the group values of enthalpy of formation, standard entropy, and heat capacity, and they are observed to be considerable for certain groups, especially for the enthalpy of formation values of OO/C/H and its adjacent groups. It is also found that the correct and consistent application of optical isomer corrections in the group additivity method is essential for accurate model predictions. The high optimization potentials of a number of groups motivate the extension of an established kinetic model optimization framework as part of this work, calibrating jointly the rate rules and group values against sets of both global model prediction targets and known species thermochemical properties. This methodology is applied to the considered 2-MH model, yielding very good posterior predictions of ignition characteristics. Overall, the results of this study demonstrate that uncertainties in certain group values will strongly propagate into the model prediction, which motivates future research towards a reduction of those uncertainties.

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## 1. Introduction

The development of chemical kinetic models has greatly improved the understanding of the oxidation mechanisms of hydrocarbon fuels and allows for the modeling of real fuel combustion processes in computational fluid dynamics simulations [1,2]. The model prediction accuracy depends on the one hand upon the validity of the kinetic mechanism itself and on the other hand heavily upon the underlying model parameters, i.e., the kinetic, thermochemical, and transport parameters. For good model predictions, accurate values of these model parameters are therefore desired. In particular, uncertainties in model parameters can propagate into

the model response, resulting in prediction uncertainties in quantities of interest [3].

Many modeling and uncertainty quantification studies have focused on the impact of kinetic rate parameters and on improving their accuracy. Often, the impact of small changes in the different rate parameters on the resulting model predictions at a specific condition is determined during the model development process by means of sensitivity and uncertainty analysis based on local or global methods [3–6]. This allows for the identification of those important reactions for which more accurate rate data would be of highest practical value, as they would considerably reduce the model prediction uncertainties for the quantities of interest. The reaction rates for elementary reactions of small fuels are often determined from both fundamental experiments and *ab initio* calculations [7]. As the number of important species and reactions grows rapidly with the fuel molecule size, mechanisms of

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larger hydrocarbons have been commonly developed in a hierarchical fashion based on the concept of reaction classes and rate rules [8–12]. In order to improve model prediction accuracies, Bugler et al. [13] have recently evaluated the rates of elementary reactions of same reaction classes for different fuels based on a thorough literature review, and consequently derived a set of new rate rules for the low-temperature oxidation of alkanes. The application of these rate rules has significantly enhanced predictions by accordingly updated kinetic models [13–18].

In contrast to the kinetic rate parameters, considerably less attention has been paid to the importance of thermochemical species parameters during model development [19]. Thermochemical parameters are either estimated from the group additivity method [20–22], calculated using quantum chemistry methods [23,24], or determined from fundamental experiments [25], and they are often incorporated directly in the chemical model without further modification and improvement. Nevertheless, these properties unavoidably involve uncertainties as well, which may affect the model prediction accuracy strongly [19]. For methane combustion, thermochemical parameters of few species were found to be important to predict certain quantities of interest such as maximum flame temperatures and maximum concentrations of some species [26,27], though for important global targets such as laminar flame speeds and ignition delay times, the impact of the kinetic parameters was found to be dominant [5,27,28]. It was found that thermochemistry has a considerably larger impact on model predictions for intermediate temperature ignition and oxidation of larger fuels with low-temperature chemistry [29,30], since chemical equilibria mainly determine the competition of low-temperature chain-branching channels with less reactive pathways. Note that since the backward rate coefficients are usually computed from the forward rate coefficients and thermochemistry, the impacts of thermochemical species properties on model predictions also depend on the used mechanism, where the corresponding reactions can be defined in different directions. In our previous work [31], the impact of all species thermochemical properties, i.e., enthalpies of formation, standard entropies, and heat capacities, on ignition predictions was evaluated over a wide range of conditions. The fuel, peroxy, hydroperoxide, and peroxy hydroperoxide radicals were found to be the most influential species at intermediate temperatures. This motivated the application of a kinetic model optimization methodology to constrain both these important thermochemical species parameters and the most sensitive kinetic rate parameters by a target set of ignition delay data for the example of auto-ignition of diethyl ether, resulting in considerably improved model predictions [32]. The optimization of thermochemical parameters of different species in a mechanism may become problematic for larger fuels, as it would result in a too large number of open parameters and neglect possible correlations between the thermochemical data of different intermediate species. Such correlations can originate, for instance, from the consideration of similar chemical groups in these intermediates during thermochemistry estimation, when thermochemical properties are determined from group additivity [20–22].

The group additivity method [20–22] is a frequently used approach for estimating the thermochemistry of combustion intermediates, especially for larger hydrocarbons. In the hierarchical mechanism development for such fuels, it is often applied jointly with the concept of reaction classes and rate rules, which is used for the kinetic model parameters. Several studies have focused on the improvement of group values [33–46]. For alkanes and other important species types, they have been recently updated by Burke et al. [43] based on accurate species data collected from the literature. Along with the updated rate rules [13], these updated group values [43] have improved kinetic model predictions and have at the same time considerably reduced error cancellation

in the correspondingly updated models [13–18]. However, while the shifted model responses with these new group values highlight their large general importance, the detailed impacts of different group values on model predictions have never been studied systematically.

Such information would be valuable in order to determine those groups, for which highly accurate values are most important in terms of the resulting model predictions. Furthermore, it is worthwhile to investigate to what extent the consistent use of thermochemical species data from group additivity in kinetic models reduces the prediction uncertainty space due to parameter correlations [47], compared to using independently determined species data. A similar uncertainty reduction due to parameter correlations had been observed for rate rules that are used by sets of elementary reactions [48,49].

Therefore, the present work aims to investigate the impacts of different group parameters on the ignition prediction by kinetic models. First, the local sensitivity analysis method is implemented for thermochemical groups to analyze the effect of incremental changes in specific group thermochemical parameters on global model predictions over a range of relevant conditions. As a case study, the auto-ignition of 2-methylhexane (2-MH) is investigated, employing the mechanism of Mohamed et al. [17]. The impact of non-nearest neighbor interaction groups and optical isomerism is analyzed, and the cancelation of sensitivities on thermochemical data of single species is quantified with respect to the resulting sensitivities on the underlying groups. Then, the uncertainties of the group values are estimated based on the covariances of their refitted values, which allows to determine their potential impacts on model prediction as products of group sensitivities and uncertainties. Finally, the established kinetic model optimization methodology based on rate rules [50–52] is extended towards a joint optimization of rate rules and groups against both global model prediction targets and known species thermochemical parameters, with application to the studied example of 2-MH ignition. Advantages and drawbacks are discussed and compared with the sequential approach of optimizing group values against species data alone and subsequently calibrating rate rules against global targets.

The remainder of this article is structured as follows. The main features of the prior 2-MH mechanism used for the present investigation are summarized in Section 2, along with the employed experimental targets. Thereafter, the group additivity method as well as the applied methodologies of sensitivity analysis, uncertainty analysis, and joint optimization are outlined in Section 3. Results of the sensitivity analysis are discussed in Section 4. Section 5 presents the determined group uncertainties, the resulting optimization potentials in both groups and rate rules, and the predictions by the 2-MH models with jointly and with sequentially optimized group values and rate rules, respectively. Finally, Section 6 concludes the paper.

## 2. Kinetic mechanism

The impact of group values and of their optimization is investigated in the present work in terms of the oxidation of a heptane isomer, 2-MH, for which the prior mechanism of Mohamed et al. [17] is considered. This branched alkane is chosen here as it contains the relevant carbon sites, i.e., primary, secondary, and tertiary carbon sites, and its carbon chain is long enough so as to enabling the low-temperature combustion chemistry. In addition, its kinetics and thermochemistry have been recently studied and updated with the state-of-the-art knowledge [17]. This ensures the high relevance of the major findings of this study, which are expected to be qualitatively applicable to structurally similar fuel components.

The model [17] is based on the original mechanism of Sarathy et al. [53], which has been updated [17] in terms of reaction pathways, rate rules, and thermochemical properties. New rate rules and group values, as proposed by Bugler et al. [13] and Burke et al. [43], respectively, were incorporated [17]. The prior version of the mechanism [17] without additional modifications is taken into account in the present work. Kinetic simulations are performed using the homogeneous constant volume reactor module in the FlameMaster software package [54].

### 2.1. Parameter uncertainties

The updated group values from Burke et al. [43] were used in the present work to compute the thermochemistry of fuel-specific species in the sensitivity analysis, and were then further optimized, first against species data [43] alone, and then jointly with the rate rules against both species data [43] and global prediction targets of the employed 2-MH model. The prior group values were assumed to be uniformly distributed, while estimates for the group uncertainties after optimization against species data alone [43] were derived from the covariance matrix of this fit assuming a multivariate normal distribution for the optimized parameters, as outlined in detail in the following sections. In the subsequent joint optimization of groups and rate rules, uniform prior distributions were used again for the group values without explicit specification of uncertainty limits. By still taking the species data [43] additionally into account as targets in this optimization, it was ensured that the posterior group values stay close to their prior values and in physically reasonable ranges. To limit the number of open parameters in the joint optimization, heat capacity group parameters were varied by constant temperature-independent values, which is analogous to the assumption of temperature-independent uncertainties in species heat capacities in previous work [32]. Following the recommendation of Bugler et al. [13], log-uniform prior distributions were also assumed for the pre-exponential factors of rate rules, and the reported uncertainties [13] were taken into account, if applicable. For the H atom abstraction reactions from the fuel by OH, small uncertainty factors of 1.5 were employed [17], as these rate rules have been measured accurately [55,56]. As in previous works [17,32,52,57–59], uncertainty factors of four were assumed for other rate rules in the absence of specified uncertainties, as larger uncertainties than in the underlying elementary reactions are expected. This allows for good comparability with previous rate rule optimization studies, where the same uncertainty estimates had been employed [17,58].

### 2.2. Experimental targets

To constrain the parameters by global model prediction targets, 11 first-stage and 59 total ignition delay data points from shock tube experiments with specified measurement uncertainties of 20% [17] were taken into account. These cover a wide temperature range of 684–1288 K, pressures of 20 and 40 atm, and equivalence ratios of 0.5 and 1 [17]. Ignition delays were chosen here as targets, since these typically exhibit particularly high sensitivities towards fuel-specific chemistry of large hydrocarbons at intermediate and low temperatures [60]. If quantitative speciation data of intermediate species were available for the oxidation of 2-MH, these could serve as interesting data for model validation and sensitivity testing as well.

## 3. Methodology

This section describes the group additivity method and the employed methodologies of sensitivity analysis, uncertainty analysis, and joint optimization of group values and rate rules.

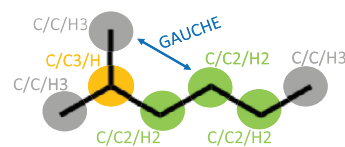


Fig. 1. Groups of the 2-MH molecule.

### 3.1. Group additivity method

While different group contribution methods are available [61–64], the group additivity method of Benson and co-workers [20–22] is most commonly used for the estimation of thermochemical species properties in kinetic models. It divides each species into a set of constituent groups, of which each represents a polyvalent atom with its ligands [20]. In addition, it also considers some important non-nearest neighbor interactions through additional correction groups [65].

A number of numerical codes [66–69] have been developed for thermochemical property estimation based on group additivity, distinguishing themselves slightly by group definition based on user-provided species descriptors. In this work, we follow the implementation of the THERM program [66,67], which has been employed in many recent kinetic modeling studies [13–18]. The group notation as introduced in the context of THERM [66,67] is thus used hereafter. An example is given in Fig. 1, which depicts the groups of the 2-MH molecule. It contains one tertiary (C/C3/H), three secondary (C/C2/H2), and three primary (C/C/H3) carbon groups as well as a GAUCHE correction group.

Typically, species values are calculated based on group additivity for the enthalpy of formation and the standard entropy (both at 298 K) as well as for the heat capacity at a set of discrete temperatures. For a molecule  $i$ , these thermochemical species values  $\varphi_i$  ( $\varphi_i = \Delta h_{f,i}(298\text{ K}), s_{\text{int},i}^0(298\text{ K}), c_{p,i}(T)$ ) are thus determined based on the values  $\psi_j$  of the underlying groups ( $\psi_j = \Delta h_{f,j}(298\text{ K}), s_j^0(298\text{ K}), c_{p,j}(T)$ ) as

$$\varphi_i(\psi) = \sum_{j=1}^m a_{ij} \psi_j, \quad (1)$$

where  $a_{ij}$  is the number of groups  $j$  in species  $i$  and  $s_{\text{int},i}^0(298\text{ K})$  refers to the intrinsic standard entropy [65]. For the standard entropy, mixing effects due to symmetry and optical isomerism need to be taken into account additionally, so that it can be computed from the intrinsic standard entropy as [65]

$$s_i^0(298\text{ K}) = s_{\text{int},i}^0(298\text{ K}) - R \ln(\sigma_i/n_{\text{OI},i}), \quad (2)$$

where  $\sigma_i$  is the symmetry number and  $n_{\text{OI},i}$  the number of optical isomers of the species. In the implementation used by THERM [66,67] and in the present work, the optical isomer contribution is accounted for by means of adding one “OI” group with an entropy group value of  $R \ln(2)$  for every chiral center in the species. Besides, OI groups are also added for OO and OOH groups in peroxy and hydroperoxide species, which are commonly regarded as pseudochiral centers [70].

For the group additivity based estimation of radical thermochemistry, two approaches exist. In the original approach [22], separate groups are used for radical-centered and radical-adjacent atoms instead of the groups used for the respective parent molecules. As an alternative, Lay et al. [71] proposed the hydrogen bond increment (HBI) approach, which is also implemented in THERM [66,67] and used in the present work. This method computes radical thermochemistry based on the thermochemistry of the corresponding parent molecule by adding a so-called bond dissociation (BD) group, that accounts for the difference in thermochemistry between the radical and its parent due to the broken

hydrogen bond. The radical enthalpy of formation  $\Delta h_{f,i}^{\text{radical}}$  (298 K) is thus calculated as

$$\begin{aligned} \Delta h_{f,i}^{\text{radical}}(298 \text{ K}) \\ = \Delta h_{f,i}^{\text{parent}}(298 \text{ K}) + \Delta h_{f,i}^{\text{BD}}(298 \text{ K}) - \Delta h_f^{\text{H}}(298 \text{ K}), \end{aligned} \quad (3)$$

where  $\Delta h_{f,i}^{\text{parent}}(298 \text{ K})$  is the parent molecule's enthalpy of formation,  $\Delta h_{f,i}^{\text{BD}}(298 \text{ K})$  is the BD group value, and  $\Delta h_f^{\text{H}}(298 \text{ K}) = 217.998 \text{ kJ/mol}$  the enthalpy of formation of the abstracted hydrogen atom. For the radical standard entropy and heat capacity, the BD group values are simply added to the parent molecule's values, i.e.,

$$\begin{aligned} s_{\text{int},i}^{0,\text{radical}}(298 \text{ K}) &= s_{\text{int},i}^{0,\text{parent}}(298 \text{ K}) + s_i^{0,\text{BD}}(298 \text{ K}), \\ c_{p,i}^{\text{radical}}(T) &= c_{p,i}^{\text{parent}}(T) + c_{p,i}^{\text{BD}}(T). \end{aligned} \quad (4)$$

Note that symmetry number and number of optical isomers may be different for a radical and its parent molecule, which has to be considered for calculation of the radical entropy.

For application in kinetic models, the derived species enthalpies of formation, standard entropies, and heat capacities are then typically expressed in the form of the well-known NASA polynomials through a further fitting step [72]. In the present work, the impact of this fitting procedure on model predictions was examined by varying the fitting hyperparameters, such as the break-point temperature between the low- and high-temperature polynomials [72], within typical ranges. However, the resulting differences in model predictions were found to be marginal for the conditions investigated here.

### 3.2. Sensitivity analysis on groups

Local sensitivity analyses on rate coefficients are commonly performed during kinetic model development to identify those reactions whose kinetic parameters have the largest impact on model predictions. They have also been applied to determine the most relevant species thermochemical parameters in terms of ignition prediction [27,31,32]. Following these works, local sensitivity analyses are used in the present study to quantitatively evaluate the impact of the different group thermochemical parameters on ignition prediction. Non-dimensionalized sensitivity coefficients of ignition delay time with respect to group parameters  $\psi_j$  are thus defined as

$$SC_j^{\psi} = \frac{\partial \tau_{\text{ig}}}{\partial \psi_j} \frac{\psi_{\text{norm}}}{\tau_{\text{ig},0}}, \quad (5)$$

where  $\psi_{\text{norm}}$  is a non-dimensionalization value, that is chosen to be  $R \cdot 298 \text{ K}$  for sensitivities on enthalpy group parameters and  $R$  for sensitivities on both entropy and heat capacity group parameters in the present study. Note that sensitivities on rate coefficients are usually normalized by the nominal rate coefficients [3], which can vary over several orders of magnitude, yet never become negative. In contrast, group values vary over a much smaller absolute range, but can take negative values. Thus, the nominal group values are not used for normalization here. For convenience, sensitivity coefficients for enthalpies of formation, standard entropies, and heat capacities will later be denoted by  $SC_j^h$ ,  $SC_j^s$ , and  $SC_j^{cp}$ . They are computed based on a brute-force approach: The value of  $\Delta h_f(298 \text{ K})$ ,  $s^0(298 \text{ K})$ , or  $c_p$  of a particular group is first altered by a pre-defined incremental value. The thermochemical properties of all relevant species are then recalculated using the altered group value, and new NASA polynomials are generated for these species. Based on them, the targets of interest, e.g., ignition delay times in the present work, are recomputed and consequently, the sensitivity coefficients can be determined. Analogous to the temperature-independent variation of heat capacity values during mechanism

optimization (compare Section 2.1), the sensitivities on heat capacity are defined such that they represent the change in model response for an equal increase of heat capacity group values at all discrete points for which group values are defined, independent of temperature.

While first-order local sensitivity coefficients, as computed here, are very useful in identifying the most impactful parameters efficiently, they cannot fully capture the actual non-linearity of sensitivities within the uncertainty intervals of the parameters. More accurate estimates of the relative parameter impacts on model predictions, taking into account these non-linearities, could be obtained via global sensitivity analysis methods [3]. Note that second-order response surfaces are used later in the optimization, thus accounting for potential non-linear effects.

### 3.3. Optimization and uncertainty estimation of group values based on species data

Various studies [33–46] have been dedicated to the extension of the set of groups proposed by Benson [22] and the refinement of their values or the development of new consistent sets of group values. The values are typically determined by means of optimization against the well-quantified thermochemical data of a set of species in a hierarchical way. For instance, in the recent work of Burke et al. [43], all alkane groups have been optimized first and have then been kept constant during the subsequent optimization steps of those group values belonging to alkenes, alcohols, ketones, and hydroperoxides [43].

The standard deviations among the species target data from the different evaluated literature studies were reported as well [43]. These vary considerably between different species, suggesting different levels of confidence in these data used to fit the group values, which was however not considered in [43]. In order to investigate the effect of considering these uncertainties as weighting factors in the optimization of group values, the group values are reoptimized by weighted least-square minimization in the present work. This also allows for estimating uncertainties in the reoptimized group values by propagating the uncertainties in the target species data back into the group values. Considering the same species data  $\varphi_i^{\text{obs}}$  as targets as in the study of Burke et al. [43] and the same hierarchical optimization approach, the group values  $\psi_j$  are determined based on the objective function

$$\Phi(\psi^*) = \min_{\psi} \sum_{i=1}^n \left( \frac{\varphi_i(\psi) - \varphi_i^{\text{obs}}}{\sigma_i^{\text{obs}}} \right)^2, \quad (6)$$

where  $\sigma_i^{\text{obs}}$  denotes the standard deviations of the species thermochemical parameters and  $\psi^*$  the vector of optimized group values. Note that the standard deviations  $\sigma_i^{\text{obs}}$  reported by Burke et al. [43] are solely based on the variances among data from different sources. The actual uncertainties in the target species data may also be higher due to systematic errors or lower due to underweighting of the most accurate values among the original sources. However, since the actual uncertainties are often missing in the literature and are difficult to quantify for the large variety of used sources, the reported standard deviations [43] are used here as estimates of the real standard deviations, assuming the target species data to be normally distributed (similar to, e. g., [73]).

A multivariate normal distribution is assumed for the optimized group values, which has been common practice in similar regression studies, for instance in the context of the Active Thermochemical Tables [73], the NEAT method [74], or the multi-scale informatics approach [75,76]. The parameter covariance matrix can then be



expressed as

$$\Sigma = \left( \sum_{i=1}^n \frac{\mathbf{a}_i \mathbf{a}_i^T}{(\sigma_i^{\text{obs}})^2} \right)^{-1} \quad (7)$$

where  $\mathbf{a}_i$  is the vector of optimized groups  $j$  contained in species  $i$ , compare Eq. (1), and has the length  $m$  of all groups considered in the optimization. The uncertainty space of  $\psi^*$  is determined by taking the Cholesky factorization of  $\Sigma$ , and the marginal standard errors are given as the diagonal entries of the Cholesky matrix. Following the common practice of uncertainty quantification for thermochemical properties [77], parametric uncertainties  $\Delta\psi_j$  are reported in the following in terms of  $2\sigma$  limits. Note that this evaluation only considers parametric uncertainties in group values propagated inversely from the underlying species data. Structural uncertainties in the group additivity model itself, i.e., residual errors from using the group additivity approximation to estimate species thermochemistry [47] and from the treatment of symmetry and optical isomer corrections [65], are beyond the present work's scope.

### 3.4. Optimization potentials

While the analysis of group sensitivities gives insight into the importance of different groups for model predictions, it does not provide full information on which groups require further improvement or optimization the most in terms of the resulting model prediction uncertainties. A group value with high sensitivity but very low uncertainty will be relatively uncritical, and vice versa. In order to assess parameter sensitivities and uncertainties jointly, local “impact factors” [78,79] have been employed for elementary reaction rates. Analogously, “optimization potentials” have been defined as products of parameter sensitivities and uncertainties in the context of rate rule and thermochemistry optimization [31,52]. These also allow for a quantitative comparison of the importance of different thermochemical parameters, such as enthalpies of formation and standard entropies, as well as a comparison with the rate parameters. Following the previous studies [31,52], the optimization potential of the group parameter  $\psi_j$  is obtained as

$$\text{OP}_j^\psi = \text{SC}_j^\psi \cdot \frac{\Delta\psi_j}{\psi_{\text{norm}}} \quad (8)$$

High optimization potentials suggest the further need for improvement of parameters. In the case of group values, one possibility is the incorporation of more accurate species data for reoptimization of the group values. Another option is the optimization against global kinetic model validation targets such as ignition delay times, motivated by previous optimization studies on chemical mechanisms [50–52,57,58,80–87]. In the literature, group values have typically been determined first by optimization against species data [43] and have then been kept constant, while automatic model optimization methodologies have been applied to calibrate the rate rules for improved model prediction accuracy [17,57,58]. However, it will be demonstrated in the following sections that the remaining optimization potentials after group optimization against species data alone can be considerable for a number of groups, and the optimization of these group values against species data and additional experimental model prediction targets thus promises a further reduction of their uncertainties. This motivates the extension of the established kinetic model optimization methodology [50–52,80] towards the joint optimization of rate rules and group values in the present work, analogous to the joint optimization of reaction rates and species thermochemical parameters in the case where thermochemistry is not based on group additivity [32].

### 3.5. Joint optimization of group values and rate rules

The determined group values, as described in Section 3.3, are taken as nominal values for the prior model. As commonly [50,51], pre-exponential factors of rate rules and group values are expressed in their normalized forms

$$x_j^A = \ln \left( \frac{A_j}{A_{j,0}} \right) / \ln f_j \quad \text{and} \quad x_j^\psi = \frac{\psi_j - \psi_{j,0}}{\Delta\psi_j} \quad (9)$$

where  $f_j$  is the uncertainty factor of rate rule  $j$ . As outlined previously in Section 2.1, the group values are treated as uniformly distributed without applying explicit uncertainty limits, and the thermochemical data of the target species are considered as optimization targets in addition to global combustion quantities of the model, such as ignition delay times. This ensures that group values stay in chemically reasonable regions.

Hence, model parameters are calibrated in the joint optimization procedure against  $R$  global prediction targets and  $n$  thermochemical parameters of target species by minimizing the objective function

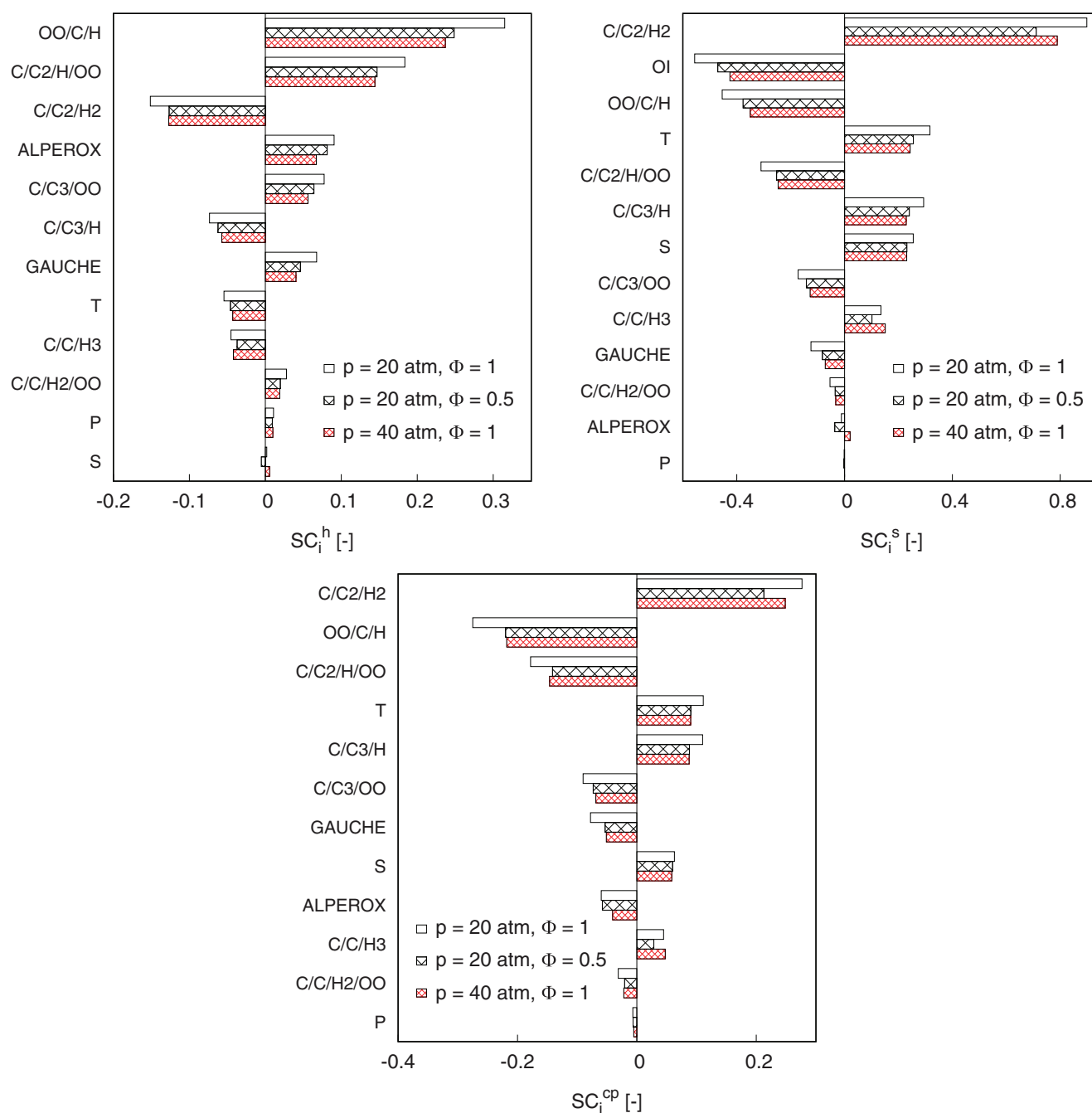
$$\Phi(\mathbf{x}^*) = \min_{\mathbf{x}} \left[ \sum_{r=1}^R \left( \frac{\eta_r(\mathbf{x}) - \eta_r^{\text{obs}}}{\sigma_r^{\text{obs}}} \right)^2 + \sum_{i=1}^n \left( \frac{\varphi_i(\mathbf{x}) - \varphi_i^{\text{obs}}}{\sigma_i^{\text{obs}}} \right)^2 \right] \quad (10)$$

where  $\eta_r^{\text{obs}}$  denotes the experimental target values and  $\sigma_r^{\text{obs}}$  the corresponding experimental standard deviations. The kinetic model predictions  $\eta_r(\mathbf{x})$  are expressed as second-order response surfaces based on the sensitivity analysis based method [88], where the sensitivities for group values are determined in the same general fashion as described in Section 3.2. Note that the thermochemical species parameters  $\varphi_i$  in the objective function include all thermochemical parameters, i. e., enthalpy of formation at 298 K, standard entropy at 298 K, and heat capacity, for each relevant species. It will be shown in the following section that only alkane and hydroperoxide groups are considerably sensitive for the investigated case of 2-MH oxidation, and only the corresponding alkane and hydroperoxide species data from [43] constraining these groups are thus considered as targets in the present work. An overview of these alkane and hydroperoxide groups as well as the corresponding target species is given in Tables S1 and S2 of the Supplementary Material (SM).

Note that the individual experimental targets and species targets are weighted equally in the objective function (Eq. (10)), following the approach of Sheen and Wang [51]. Weightings for these two different types of targets could be refined, for instance, if the number of experimental target data were considerably larger than the number of species target data, and a higher weighting of species data were desired. In such a case, the two datasets could be assigned different weighting factors in the objective function, similar to the weighting of different experimental datasets proposed by Turányi et al. [82]. Also, by assuming uniformly distributed prior parameters, the objective function favors strong modifications of parameters. Alternatively, if the prior parameters were assumed to be normally distributed, additional regularization terms in the objective function would serve as penalties to prevent strong modifications of individual parameters [51]. Following the recommendation of Bugler et al. [13], uniform prior distributions are considered in the present study, as outlined earlier.

## 4. Sensitivity analysis of the prior model

This section presents results of the sensitivity analysis for ignition delay times on groups. The most influential groups are identified over a wide range of temperatures, and the importance of



**Fig. 2.** Sensitivity coefficients of ignition delay time on the group values of  $\Delta h_f(298\text{ K})$ ,  $s^0(298\text{ K})$ , and  $c_p$  for a 2-MH/air mixture at 870 K.

gauche interaction groups and optical isomer corrections is analyzed and discussed in detail. In addition, the mutual cancelation of sensitivities on single species thermochemical properties in the resulting sensitivities on their underlying group values is quantified.

#### 4.1. Sensitivities on group values

The ranked sensitivity coefficients of ignition delay time with respect to the group values of  $\Delta h_f(298\text{ K})$ ,  $s^0(298\text{ K})$ , and  $c_p$  are depicted in Fig. 2 for a temperature of 870 K and three different conditions of pressure and equivalence ratio. These conditions lie well within the negative temperature coefficient (NTC) regime, where both low and high temperature reaction channels are active. Igni-

tion delay times are very sensitive towards a number of groups at these conditions, which is expected from the observed high sensitivities on single species thermochemical properties at intermediate temperatures [31]. Since enthalpies of formation, standard entropies, and heat capacities all affect mainly the chemical equilibria of important elementary reactions [31], important groups are the same in all three cases. A high sensitivity on standard entropy is observed for the optical isomer correction (OI) group. This is greatly due to the presence of pseudochiral centers in peroxy, alkyl hydroperoxide, and peroxy hydroperoxide species, for which OI groups are added, as mentioned earlier. The competition of their  $\text{O}_2$  addition reactions with less reactive channels leads to high sensitivities on thermochemistry in the NTC regime, and thus also with respect to the groups these species comprise. Note that the

high sensitivity with respect to the OI group has zero impact on the model prediction uncertainties. The corresponding correction value to the entropy of  $R\ln(2)$  for every chiral center in a species is derived directly from the thermodynamic expression for the entropy of mixing [65] and is an inherent part of Benson's group additivity method without parametric uncertainty. Nevertheless, the high sensitivity indicates the importance of the correct application of the OI group for the estimation of thermochemical properties, which will be discussed in more detail in Section 4.3.

The other groups with high sensitivities include mainly those which are part of the main intermediate species of the low-temperature chain-branching pathways. For instance, the groups OO/C/H as well as C/C2/H/OO, C/C/H2/OO, and C/C3/OO represent the OO and OOH moieties as well as their adjacent groups in peroxy, alkyl hydroperoxide, and peroxy hydroperoxide radicals. Interestingly, at these conditions, sensitivities with respect to these groups are all positive for enthalpies of formation, while they are all negative for standard entropies and heat capacities. In both the first and second  $O_2$  addition reactions, the product species (i.e.,  $RO_2$  and  $OOQOOH$ , respectively) have one OO moiety more than the educt species (i.e., fuel radical and  $QOOH$ , respectively). Therefore, increasing the enthalpy of formation value of one of the aforementioned groups will lead to a stronger increase in species enthalpy of formation and chemical potential for  $RO_2/OOQOOH$  than for fuel radical/ $QOOH$ . This shifts the equilibrium of the  $O_2$  addition reactions backwards, resulting in decreased reactivity, higher ignition delays, and therefore positive sensitivities. The contrary effect of standard entropies and heat capacities, which exhibit negative sensitivities with respect to these groups compared to the positive sensitivities on enthalpies of formation, is due to their reverse impact on the species chemical potentials, which ultimately determine chemical equilibria [31]. Other relevant groups in Fig. 2 include the BD groups ALPEROX (for peroxy radicals), P (for primary radicals), S (for secondary radicals), and T (for tertiary radicals), which need to be considered for R,  $RO_2$ ,  $QOOH$ , and  $OOQOOH$  radicals, as well as the primary (C/C/H3), secondary (C/C2/H2), and tertiary (C/C3/H) carbon groups of the basic fuel structure (see Fig. 1).

The evolution of group sensitivities over a wide temperature range is depicted for enthalpies of formation in Fig. 3, while similar representations for standard entropies and heat capacities can be found in Figs. S1 and S2 in the SM. As discussed before, sensitivities peak at intermediate temperatures around 900 K. At higher temperatures, only the fuel and its radicals are moderately sensitive among the fuel-specific species in terms of ignition delay predictions, which was confirmed by sensitivity analyses on elementary reactions and species thermochemical properties. For the fuel molecule itself, this sensitivity is largely due to thermal effects rather than shifted equilibria [31]. Consequently, only the primary, secondary, and tertiary carbon groups C/C/H3, C/C2/H2, and C/C3/H, which are the basic structural constituents of the fuel and its radicals (see Fig. 1), are moderately sensitive at high temperature.

At very low temperatures, the sensitivities on the group parameters related to the peroxy and hydroperoxide moieties (OO/C/H and C/C2/H/OO) become negative. This trend correlates with negative net sensitivities on  $OOQOOH$  at these temperatures, which were observed in additional sensitivity analyses on species enthalpies of formation, compare Fig. S3 of the SM. Normally, an equilibrium shift of the second  $O_2$  addition reactions towards  $QOOH$  would inhibit reactivity due to reduced chain branching. However, the main contributing  $OOQOOH$  isomer to the net sensitivity on  $OOQOOH$  in this region turns out to be *c7ooh2-oo4-2* (see [17] for nomenclature), which is produced through the channel starting from the tertiary fuel radical. As no ketohydroperoxides can be produced from this pathway [17],  $OOQOOH$  can only pro-

ceed via alternative isomerization to the corresponding  $P(OOH)_2$  isomer, which would further react into hydroperoxide cyclic ether. This pathway is less reactive than the reaction of the preceding  $QOOH$  radical into cyclic ether, which was confirmed by a sensitivity analysis on elementary reactions at the same conditions. Hence, the impact of the mentioned group parameters on ignition is reversed at very low temperatures compared with intermediate to high temperatures.

For the analysis of group effects in the present study, as presented before, only the fuel-specific intermediate species have been computed from group additivity during the sensitivity analysis and optimization. This analysis captures the main important effects for model development of large hydrocarbon fuels, where typically only the fuel-specific chemistry is added upon a well-validated base mechanism, as is the case for the 2-MH model considered here [17]. However, it is worthwhile to investigate whether the sensitivities on groups change significantly if the thermochemistry of other important species is estimated based on group additivity as well. For this purpose, the most important non-fuel-specific species for ignition delay predictions of 2-MH/air mixtures in the NTC regime have been determined by sensitivity analysis on species enthalpies of formation. These species have then also been estimated from group additivity, and the sensitivity analysis on groups has been repeated, with results being shown in detail in Figs. S4 and S5 of the SM. Only small changes in sensitivities are observed compared to the case where the non-fuel-specific species are neglected, and the drawn conclusions are thus the same in both cases.

#### 4.2. Effects of parameter correlations

The afore-discussed results reveal moderate sensitivities for certain alkane molecule groups, especially for C/C/H3, whose sensitivities are considerably lower than sensitivities for the peroxy group OO/C/H, even though it is also contained in the intermediate species of the low-temperature chain branching pathways. The reason lies in strong resulting correlations among species thermochemical properties due to the consistent use of these groups. In particular, alkane molecule groups are often present in product and educt species of important reactions with the same numbers, as shown in Fig. 4 exemplarily for a low-temperature  $O_2$  addition reaction of 2-MH hydroperoxide radical.

For these reactions, the thermochemical properties of reactants and products are changed by the same absolute values, once these group values are modified. Thus, the resulting chemical equilibria are often not affected, leading to relatively small sensitivities on these groups. To quantify the extent of this compensating effect for each group, we define the "sensitivity cancellation" of group sensitivity as

$$\text{Sensitivity Cancellation}_j = \left( \sum_{i=1}^M a_{ij} |SC_i^\varphi| \right) - |SC_j^\psi|, \quad (11)$$

where  $M$  is the number of species  $i$  in the model which comprise the group  $j$ , and  $SC_i^\varphi$  and  $SC_j^\psi$  are the sensitivity coefficients on species and group thermochemical parameters, respectively. Results are depicted in Fig. 5 for sensitivities on enthalpies of formation at the same conditions as in Fig. 3. The compensating effects are found to be strongest for the C/C/H3 and C/C2/H2 groups at intermediate temperatures, whose sensitivity cancellations are multiple times higher than their actual sensitivities (see Fig. 3). In contrast, the compensating effects are relatively weak for the BD groups and the hydroperoxide groups such as OO/C/H, whose number on the product side of low-temperature  $O_2$  addition reactions is consistently higher than on the educt side.

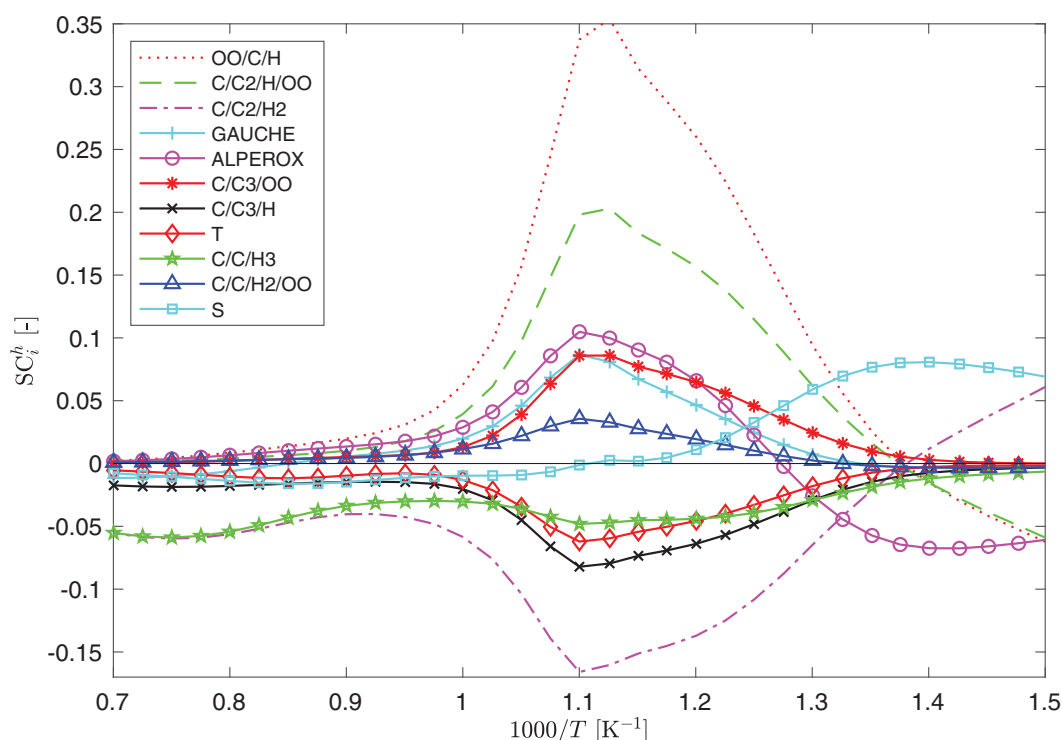


Fig. 3. Sensitivity coefficients of ignition delay time on the group values of  $\Delta h_f(298\text{ K})$  for a stoichiometric 2-MH/air mixture at 20 atm.

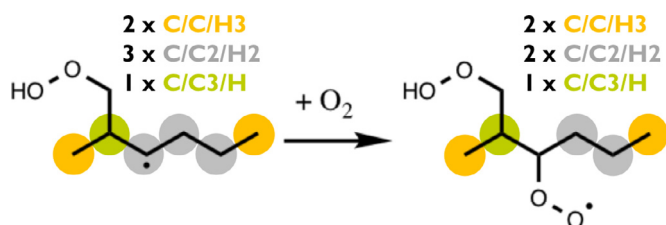


Fig. 4. Sensitivity cancellations for the alkane molecule groups in an exemplary low-temperature  $\text{O}_2$  addition reaction. The shown reaction does not contribute to sensitivities on the groups C/C/H3 and C/C3/H, as these are contained with the same numbers in the product and educt species, respectively. In contrast, the reaction partly contributes to the sensitivity on the C/C2/H2 group, whose number in the OOQOOH radical is lower than in the QOOH radical.

Overall, these results in conjunction with those presented earlier suggest that only a subset of groups is actually relevant in terms of their potentials to induce considerable uncertainties in model predictions, especially OO/C/H and its adjacent groups in peroxy species for the present case of a branched alkane fuel. Accurate values for these groups are therefore of particular importance. Other groups, however, such as the alkane molecule group C/C/H3 in the present case, are relatively less critical for accurate model predictions despite their occurrence in many important species. The consistent application of the group additivity method for all low-temperature intermediate species therefore effectively reduces the overall uncertainty space of model predictions with regards to thermochemistry, as the obtained species thermochemical properties are strongly correlated.

#### 4.3. Impact of gauche correction groups

The gauche correction groups are added in the group additivity method to account for the enthalpy effect of 1,4 (i.e., gauche) interactions between non-nearest neighbor groups in alkanes [20], as

depicted in Fig. 1. In addition, gauche interactions are also present in alkyl radicals. To account for the differences of these gauche interactions in radicals compared to “alkane gauche” (AG) interactions, separate interaction groups have been proposed depending on the location of the radical site, namely “radical gauche 1” (RG1) and “radical gauche 2” (RG2) [35], as illustrated in Fig. 6.

Different values have been proposed for these correction groups: Marsi et al. [35] found that the effect of RG2 is negligible, and they proposed a relatively small value of  $-0.2\text{ kcal/mol}$  for RG1, compared to the value of  $0.8\text{ kcal/mol}$  for AG. Sabbe et al. [40] published slightly higher values for both RG1 and RG2. Mohamed et al. [17] have used  $0.8\text{ kcal/mol}$  for both AG and RG1 in their work on 2-MH, while neglecting RG2 as suggested by Marsi et al. [35]. Note that these published gauche group values are also dependent on other group values used in the same respective literature works, with which they were fitted simultaneously.

In the present study, we aim to shed light on how sensitive the AG, RG1, and RG2 groups are in terms of ignition predictions. Two cases are compared in Fig. 7. The upper sensitivity corresponds to a case where only one gauche correction group is considered, termed GAUCHE, which was added for both AG and RG1 interactions. This corresponds to the assignment chosen during development of the employed 2-MH model [17], and the sensitivity value is equivalent to that already depicted in Fig. 2. The second case considers the groups for AG, RG1, and RG2 interactions separately during sensitivity analysis, though their nominal values are still equal to those used in the first case ( $0.8\text{ kcal/mol}$  for AG and RG1,  $0\text{ kcal/mol}$  for RG2). Clearly, the sensitivities in the second case are larger in absolute terms, as they partly compensate each other when one unified GAUCHE group is used. While the impacts of these group values on ignition delay times are only moderate in absolute terms compared to the impacts of other important groups (see previous subsection), they are overall not negligible and thus, the correct application of gauche interaction groups deserves particular attention in the estimation of thermochemical properties.



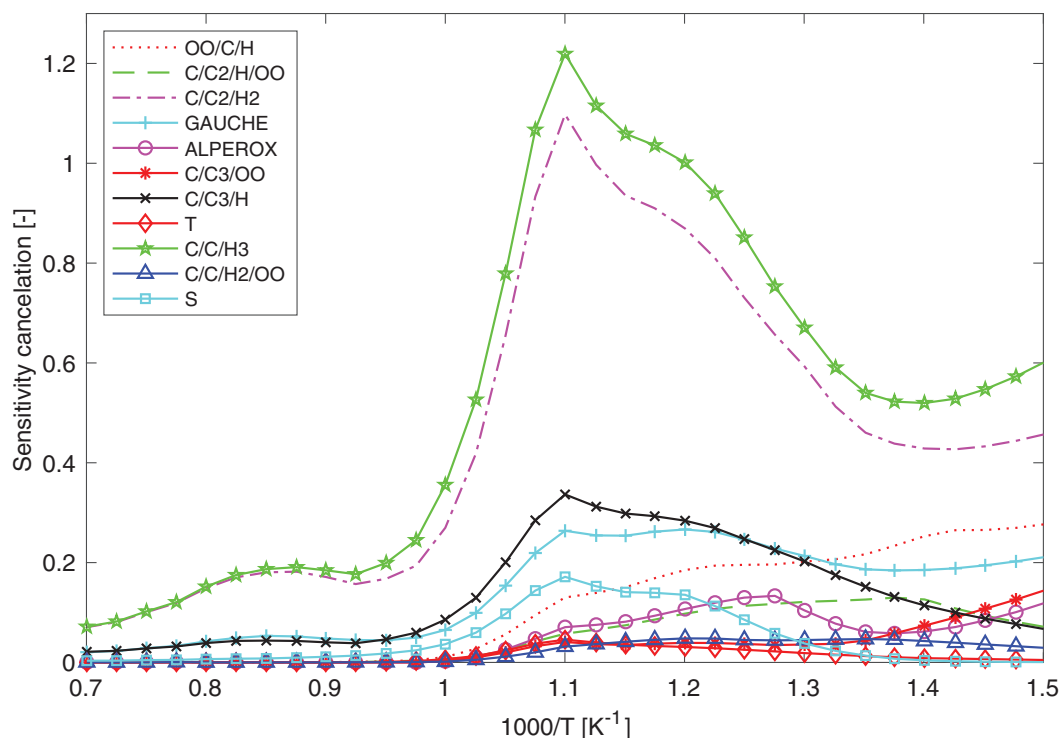


Fig. 5. Cancellation of single species sensitivities in resulting group sensitivities on enthalpies of formation for a stoichiometric 2-MH/air mixture at 20 atm.

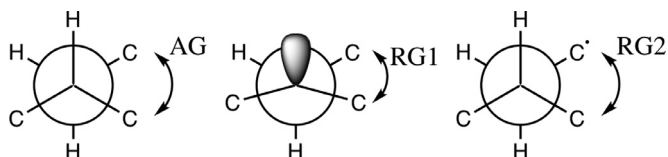


Fig. 6. Newman projections of alkane gauche (AG), radical gauche 1 (RG1), and radical gauche 2 (RG2) interactions [35], illustrated for the 2–3 bond of the gauche structure in 2-MH (see Fig. 1).

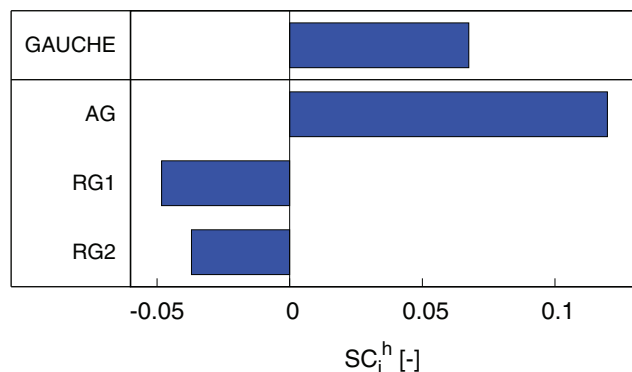


Fig. 7. Sensitivity coefficients of ignition delay time on  $\Delta h_f(298\text{ K})$  group values for the gauche interaction groups (20 atm, 870 K,  $\Phi = 1$ ). GAUCHE (upper part) represents a case, where alkane gauche and radical gauche 1 are both considered as GAUCHE groups, while radical gauche 2 is neglected. AG/RG1/RG2 (lower part) represents a case, where the three gauche correction groups are considered separately in the sensitivity analysis, while having the same nominal values as in the first case.

#### 4.4. Impact of optical isomerism

As outlined earlier, OI corrections to the standard entropy have to be added for those species which comprise chiral centers in order to account for the entropy of mixing effect arising from the optical isomers of that species, and they are also commonly added

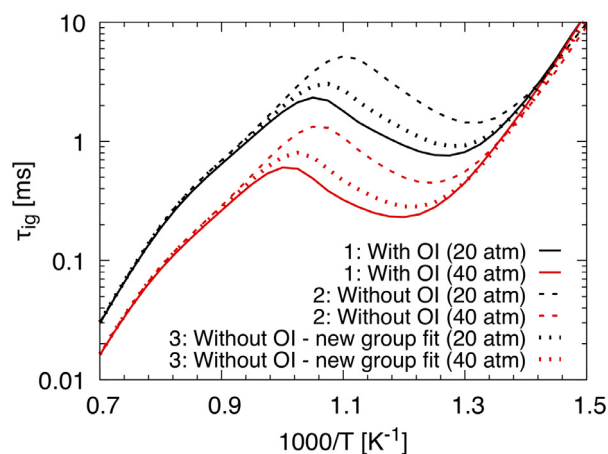


Fig. 8. Model predictions ( $\Phi = 1$ ) with (1) and without (2) taking OI groups into account. In the third shown case (3), the other group values have first been reoptimized against the target species of Burke et al. [43] by also neglecting the OI groups in these target species.

for peroxy and hydroperoxide groups, which are considered pseudochiral centers [70]. As shown in Section 4.1, the OI group is very sensitive. While its uncertainty is zero by definition, an incorrect application of the OI group may result in large deviations of model predictions. Therefore, the effect of neglecting optical isomerism completely is analyzed here as an extreme case. Figure 8 shows ignition delay times of 2-MH for one case, where the OI groups are considered in the conventional way, and for a second case, where these are neglected. The neglect leads to a strong increase in reactivity in the NTC regime, which highlights their large impact.

However, the consistent application of OI groups is already important during the development of group values based on known species values. In these respects, the OI groups for chiral and pseudochiral centers had been consistently taken into account by

Burke et al. [43] during the optimization of the other group values used here. In order to assess the effect of these being consistently neglected during both determination of group values and subsequent species thermochemistry estimation based on the optimized groups, a third case is considered here. All group values are refitted first against the species data of Burke et al. [43] by neglecting all OI groups in the target species. Note that for this particular case, unweighted least-square minimization is applied for the group optimization, in order to allow for good comparability to the other two cases in Fig. 8, where the groups of Burke et al. [43] are used. These refitted values are then employed to reestimate species thermochemistry in the 2-MH model by still neglecting all OI groups for species in the model. The difference to the first case is smaller now due to error compensation, but still considerable, see Fig. 8. These results highlight the large impact of optical isomer corrections on model predictions. Considering the overall high sensitivities of ignition delay times on OI corrections, their correct and consistent incorporation during both group value determination and species thermochemistry estimation can be more important in terms of model predictions than accurate values for a number of other groups.

## 5. Uncertainty analysis and optimization

The first part of this section presents the uncertainties in the relevant group values, which were reoptimized against the species data that had been considered by Burke et al. [43]. The remaining optimization potentials in terms of the 2-MH model are subsequently presented and compared with those of the relevant rate rules. Thereafter, the proposed joint optimization methodology is applied to jointly optimize groups and rate rules against both global prediction targets of the considered 2-MH model and additional species thermochemical data [43]. The resulting parameters and posterior model predictions are compared with the established sequential approach of first optimizing group values against species data alone and subsequently optimizing rate rules against global quantities of interest, supplemented by a discussion of advantages and limitations of both approaches.

### 5.1. Uncertainties and optimization potentials of group values

The groups were reoptimized against the species data [43] by weighted least-square minimization, as outlined in Section 3.3. With respect to the employed set of target species [43], the hydroperoxide groups OO/C/H, C/C2/H/OO, and C/C/H2/OO are linearly dependent, because OO/C/H always appears together with either of the other two groups in the used hydroperoxide species [43]. This linear dependency among groups of a certain species class (except for alkane groups) is a well-known inherent characteristic of the group additivity method [20,40,41]. In the literature, to remove the linear dependency, the values of one group among a set of linearly dependent groups have been typically fixed or set equal to the values of another group [20,40,41]. Hence, in order to allow for a unique solution and a non-singular covariance matrix as defined by Eq. (7), the values of C/C/H2/OO, which is the least sensitive among these three groups in Fig. 2, are fixed here at the values reported in [43]. Note that only the sums of OO/C/H and its respective adjacent groups can be determined uniquely, and the optimized values of these sums do not depend upon the choice of the specific group to be fixed. The resulting reoptimized group values, as well as the ignition delay predictions by the prior 2-MH model using both the reoptimized group values and those of Burke et al. [43], are provided in Table S3 and Fig. S6 of the SM, respectively. Interestingly, the use of the reoptimized group values alone leads to a shift of ignition delay predictions of more than 50% in the NTC regime compared to using the values of

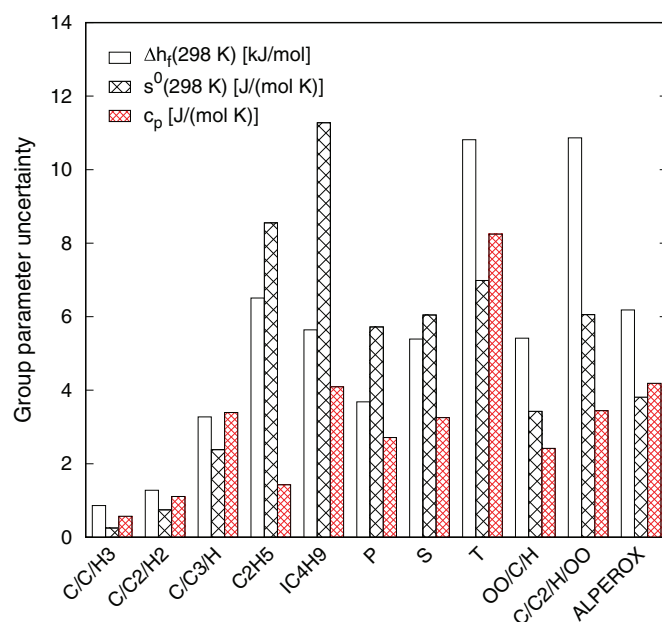


Fig. 9. Marginal uncertainties of the refitted group contribution values for enthalpy of formation, standard entropy (both at 298 K), and heat capacity. For heat capacity, the shown values represent the averages over the respective uncertainties in heat capacity group values at different discrete temperatures. The respective units of the uncertainty values are shown in the legend.

Burke et al. [43]. Thus, the model predictions already have a considerable sensitivity on the method used for derivation of group values, even if the employed dataset of target species is the same.

The marginal  $2\sigma$  uncertainties in the relevant group parameters are subsequently quantified according to Eq. (7), i. e., they are obtained from the diagonal entries of the parameter covariance matrix after group optimization against species data. Results are depicted in Fig. 9. Uncertainties are very small in the group values of C/C/H3 and C/C2/H2, which are constrained by the thermochemical data of a number of target species with low uncertainties, such as ethane, propane, n-butane, and iso-butane. For other groups, the uncertainties are larger. This can be attributed to the smaller number of species constraining the values of these groups and to the larger uncertainties in the target species data. For instance, the BD group T for tertiary radicals is constrained by only one species in the target dataset considered by Burke et al. [43], namely the tert-butyl radical, with a stated  $2\sigma$  uncertainty for enthalpy of formation of about 10.6 kJ/mol [43]. Hence, this uncertainty is directly propagated into the enthalpy of formation value of the T group, resulting in the considerable group uncertainty depicted in Fig. 9.

In order to determine those groups for which more accurate values would be of highest interest with respect to the resulting kinetic model predictions, the sensitivities have to be taken into account as well, resulting in the group optimization potentials. The absolute values of group optimization potentials are depicted in Fig. 10 for 2-MH ignition in the NTC regime, where the highest sensitivities were observed, as shown in Fig. 3. The optimization potentials are found to be largest for the enthalpy of formation values of the hydroperoxide groups C/C2/H/OO and OO/C/H as well as the BD groups ALPEROX and T, which represent peroxy and tertiary radical sites, respectively. Due to their low uncertainties and sensitivities, the alkane molecule groups, such as C/C/H3 and C/C2/H2, have relatively low optimization potentials. Therefore, from a practical kinetic modeling point of view of alkane fuels, future studies on group optimization should focus mainly on the improvement of the mentioned hydroperoxide and BD groups, for which the enthalpy of formation values are most critical.

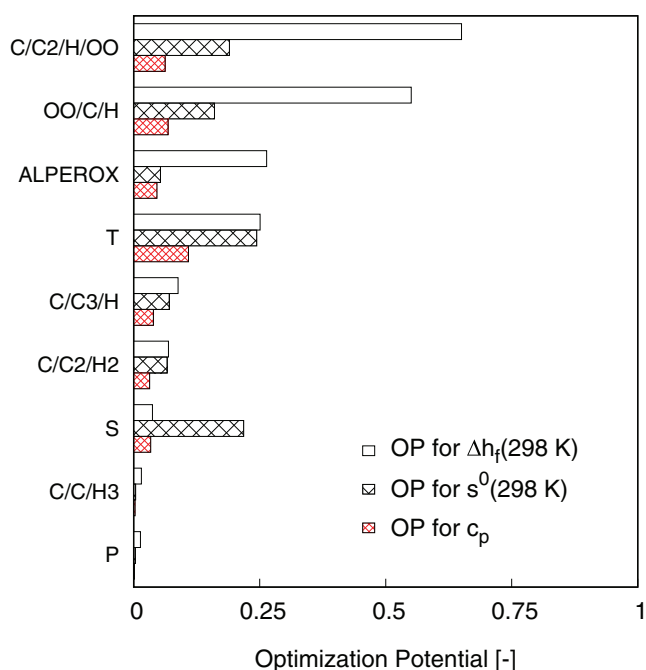


Fig. 10. Optimization potentials of group parameters for ignition delay predictions of a stoichiometric 2-MH/air mixture at 20 bar and 870 K.

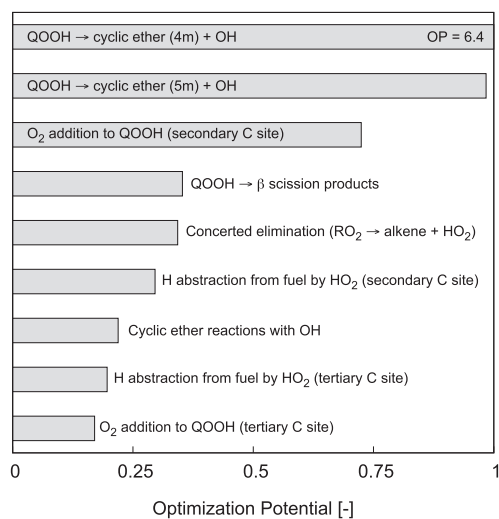


Fig. 11. Optimization potentials of rate rules for ignition delay predictions of a stoichiometric 2-MH/air mixture at 20 bar and 870 K.

## 5.2. Optimization potentials of rate rules

The results discussed so far have indicated a generally large impact of group values on kinetic model predictions. In this subsection, their relative importance to the reaction rate rules is explored. In order to allow for such comparison, optimization potentials of the 2-MH model have been determined as well for the applied rate rules, with results shown in Fig. 11 for the same condition as for the group values. Since different upper and lower uncertainty limits were provided by Bugler et al. [13] for a number of rate rules, the optimization potentials are determined here based on the respective average values of the provided upper and lower uncertainty factors. The magnitudes of optimization potentials for most rate rules lie in the same range as the magnitudes of the afore-discussed optimization potentials of groups. An exception is the rate rule for the class of QOOH reactions into cyclic ether and

OH via 4-membered rings with an optimization potential of 6.4, which is by far higher than for all other considered parameters. The reason lies in the very high reported uncertainty limits of this rate rule of 35.8 and 11.1 for the upper and lower uncertainties, respectively [13]. If the uncertainty factor for this rate rule were equal to or below 4, as assumed for most other depicted rate rules, the optimization potential would be in the range of the remaining most important rate rules and groups. Note that at high temperatures, the maximum optimization potentials of both rate rules and groups are lower due to their lower sensitivities, and the rate rules are dominant over the group values at these conditions, compare Figs. S7 and S8 in the SM. Still, the results highlight that with respect to accurate model predictions, uncertainties in a number of group values are almost as critical as uncertainties in the most important rate rules for ignition at lower temperatures.

## 5.3. Posterior parameters and model predictions of joint and sequential optimizations

The general advantages and risks of joint optimization of kinetics and thermochemistry for a reaction mechanism have been discussed in our previous work [32]. It was concluded that the optimization of the thermochemical data of single species can lead to a very high number of open parameters and to inconsistent posterior species values within a model, when it comes to larger fuels. As an alternative that avoids these drawbacks, group values are considered as open parameters and optimized jointly with the rate rules. To assess the impact on posterior model predictions of considering group values as calibration objectives in addition to rate rules, two models are developed. The first model, named “optSequential” hereafter, is based on the established concept of optimizing groups against species data alone and subsequently optimizing rate rules against global model prediction targets [52], by fixing the group values at their previously optimized values. Here, the reoptimized group values discussed in Section 5.1 are employed as the priors in the rate rule optimization. Note that in this sequential optimization, the group values are optimized first, because their optimization is independent from the used reaction model, and the rate rules are only optimized thereafter, as their optimized values depend potentially on the accuracy of the underlying thermochemical parameters in the model. The second model, named “optJoint”, is developed based on the concept proposed in Section 3.5 by jointly optimizing rate rules and group values against ignition delay data and considering the relevant species thermochemical parameters [43] as additional targets. For the sake of good comparability of the results, the same threshold of minimum optimization potentials is employed for automatic selection of the calibrated parameters. The same subset of rate rules is thus optimized in both cases.

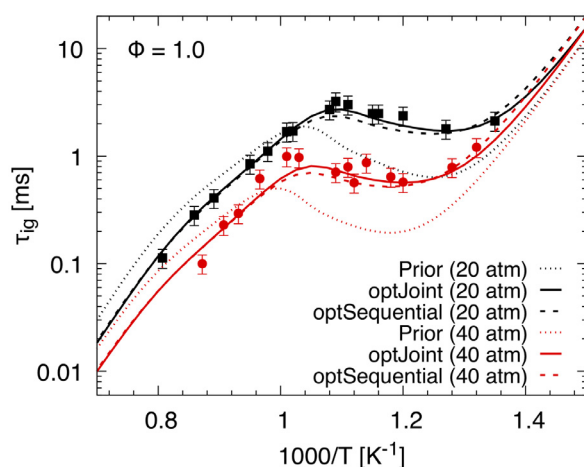
The posterior group values of the model optJoint, normalized (compare Eq. (9)) based on their marginal  $2\sigma$  uncertainty limits determined in Section 5.1, are presented in Table 1, while a list of their absolute posterior values is provided as Table S4 in the SM. For most groups, the differences of posterior and prior values do not exceed the marginal  $2\sigma$  uncertainties, except for the standard entropy values of the S and ALPEROX groups, whose changes are close to their  $2\sigma$  limits, though. Thus, the consideration of the species data [43] as additional targets in the joint optimization ensures that the group values remain in physically reasonable ranges in the present case, even though no explicit uncertainty limits are applied for the groups in the optimization. Note that in principle, the marginal  $2\sigma$  uncertainties determined in Section 5.1 could also be applied as strict uncertainty limits in the joint optimization, if strong modifications of many group values beyond their  $2\sigma$  limits were observed.

The posterior pre-exponential factors of rate rules for both cases (optSequential and optJoint) are provided in Table S5 in the

**Table 1**

Posterior normalized group values of enthalpy of formation at 298 K ( $x_i^{h,*}$ ), standard entropy at 298 K ( $x_i^{s,*}$ ), and heat capacity ( $x_i^{cp,*}$ ) of the model optJoint. The values were normalized based on the marginal  $2\sigma$  uncertainty limits determined in Section 5.1. The alkane radical groups C2H5 and IC4H9 (compare Table S1 in the SM) are not used by intermediate species in the employed mechanism, but they were included as calibration objectives in the optimization to achieve the best possible predictions of the species data [43], that were employed as additional targets.

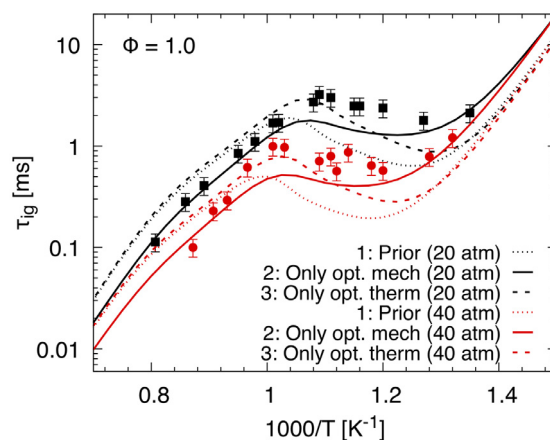
| Group     | $x_i^{h,*}$ | $x_i^{s,*}$ | $x_i^{cp,*}$ |
|-----------|-------------|-------------|--------------|
| C/C/H3    | +0.10       | +0.00       | +0.15        |
| C/C2/H2   | −0.07       | −0.28       | −0.08        |
| C/C3/H    | −0.04       | +0.05       | −0.11        |
| C2H5      | −0.03       | +0.00       | −0.12        |
| IC4H9     | −0.01       | −0.01       | +0.03        |
| P         | +0.00       | +0.04       | +0.02        |
| S         | −0.26       | +1.09       | −0.36        |
| T         | −0.65       | −0.17       | +0.29        |
| OO/C/H    | −0.09       | +0.98       | −0.02        |
| C/C2/H/OO | +0.10       | −0.15       | −0.05        |
| ALPEROX   | +0.29       | −1.02       | +0.01        |



**Fig. 12.** Ignition delay predictions by the prior model, the model with sequentially optimized groups and rate rules (optSequential), and the model with jointly optimized groups and rate rules (optJoint) for a stoichiometric 2-MH/air mixture. The symbols represent experimental target data, with the error bars corresponding to reported experimental uncertainties of 20% [17].

SM. The log-uniform prior distribution gives equal probability to the prior rate rules within their uncertainty limits, thus many values are modified by their maximum uncertainties, indicating that some rate rules would have possibly been changed even more, if higher uncertainty factors had been assumed. For a number of rate rules, the posterior values differ between the models optSequential and optJoint. Similar observations had been reported in the case of joint optimization of rates and species thermochemical parameters [32]. The reason lies in the shifted chemical equilibria of important reactions in the case of model optJoint. Due to the impact of these equilibria on reactivity, a different set of posterior values of rate rules is required to achieve accurate predictions by model optJoint, compared to model optSequential.

The model predictions of the two optimized models are compared with the experimental ignition delay data in Fig. 12 for stoichiometric conditions, while comparisons at other considered conditions can be found in Figs. S9 and S10 in the SM. Generally, good prediction accuracy is achieved by both models, with slight deviations between the two in the NTC regime. A slightly higher reactivity of model optJoint is also predicted at lower temperatures compared to model optSequential, where data for validation is missing, but the differences between both models are within typical experimental uncertainties.



**Fig. 13.** Ignition delay predictions for a stoichiometric 2-MH/air mixture compared for three cases: The prior model (1), the optimized rate rules of model optJoint with the prior groups (2), and the optimized groups of model optJoint with the prior rate rules (3). The symbols represent experimental target data, with the error bars corresponding to reported experimental uncertainties of 20% [17].

After both optimized models were found to yield good predictions of the experiments, Fig. 13 finally sheds light on the question, what the contribution of the optimized kinetics and the optimized thermochemistry is, respectively, to the decreased reactivity in case of model optJoint. Two tests are performed, one by calculating ignition delay times based on the jointly optimized rate rules and the prior groups, and the other one by using the jointly optimized groups with the prior rate rules. In the NTC regime, strongly decreased reactivities are observed for both test cases, which indicates that the modified rate rules and the modified group values both contribute strongly to the improved ignition delay predictions of model optJoint in this regime. However, while the calibration of group values improves the results only in the intermediate temperature range, the calibrated rate rules can improve the prediction over a wide range of temperatures. The reason lies in the relatively higher sensitivities of rate rules at certain conditions. Therefore, at high and low temperatures, the improvements in ignition delay time predictions are almost entirely caused by the rate rule calibration.

These results indicate that the rate rules should always be considered as calibration objectives, when a kinetic model is to be optimized, while the optimization of group values alone against ignition delay data cannot be expected to improve predictions over the same wide range of conditions. Still, the large optimization potentials of groups at intermediate temperatures motivate their inclusion in the joint optimization. However, due to the limited number of experimental targets for 2-MH as a single fuel component, the resulting system is underconstrained, as in most mechanism optimization cases [89], and the prediction improvements are almost negligible in the present work. One way to better constrain rate rules and group values jointly would be their optimization for an entire hierarchy of fuels, as previously performed for the n-alkane rate rules alone [58]. These optimized rate rules and groups could then also be applied in the kinetic mechanism development of other fuels, that were not considered as optimization targets [58]. Still, care would have to be taken in such a case when applying groups and rate rules in model development, since the optimized group values are correlated with the jointly optimized rate rules, and both sets of parameters would thus also have to be applied jointly for best prediction accuracies when developing new mechanisms. Also, the very large number of resulting open parameters can be an additional challenge for the employed optimizer in terms of finding the global optimum. Since the optimization of rate rules



alone, analogously to the sequential optimization presented here, has proven to effectively reduce prediction uncertainties and posterior parameter uncertainties as well [58], it is thus expected to be still favorable in many cases. Anyhow, the employed group values should always be carefully evaluated *a priori*, and their joint optimization with the rate rules generally appears to be a promising advancement of the established optimization concepts.

## 6. Concluding remarks

In this study, the impact of group thermochemical parameters on kinetic model predictions has been investigated for the first time systematically by means of sensitivity analysis, employing a recent 2-MH mechanism. The highest sensitivities were found in the NTC regime for OO/C/H and its adjacent groups, which correspond to the peroxy and hydroperoxide moieties of low-temperature intermediate species. Very high impact was also found for the optical isomer correction group, which highlights the importance of its correct incorporation during both group optimization and species thermochemistry estimation. On the other hand, sensitivities on certain alkane molecule groups, such as C/C/H<sub>3</sub>, were found to be only moderate due to sensitivity cancellation effects in many important reactions. Uncertainties in the group values have then been calculated by propagating uncertainties from a set of target species into the groups, and optimization potentials have been determined accordingly as products of parameter sensitivities and uncertainties. A number of hydroperoxide groups and alkane BD groups show high optimization potentials, especially with respect to their enthalpy of formation values, which motivates future work on the reduction of the associated uncertainties. The high optimization potentials of groups motivated the application of a novel methodology for joint optimization of rate rules and groups against both ignition delay times and thermochemical species data as targets. Good prediction accuracy of the experimental targets is achieved by the model using the jointly optimized groups and rate rules.

The sensitivity and uncertainty analysis approach proposed in this work helps in identifying those groups for which more accurate values would be of highest practical interest, and its future application to other fuel types could further advance this knowledge. Together with sensitivity and uncertainty analysis on rate rules, it can provide helpful guidance for kinetic modeling by identifying the most important parameters, and if necessary, allows for their further optimization.

## Declaration of Competing Interest

None.

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## Supplementary material

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.combustflame.2019.12.011](https://doi.org/10.1016/j.combustflame.2019.12.011).

## References

- [1] F. Battin-Leclerc, Detailed chemical kinetic models for the low-temperature combustion of hydrocarbons with application to gasoline and diesel fuel surrogates, *Prog. Energy Combust. Sci.* 34 (4) (2008) 440–498, doi:[10.1016/j.pecs.2007.10.002](https://doi.org/10.1016/j.pecs.2007.10.002).
- [2] C.K. Westbrook, M. Mehl, W.J. Pitz, G. Kukkadapu, S. Wagnon, K. Zhang, Multi-fuel surrogate chemical kinetic mechanisms for real world applications, *Phys. Chem. Chem. Phys.* 20 (2018) 10588–10606, doi:[10.1039/C7CP07901J](https://doi.org/10.1039/C7CP07901J).
- [3] A.S. Tomlin, The role of sensitivity and uncertainty analysis in combustion modelling, *Proc. Combust. Inst.* 34 (1) (2013) 159–176, doi:[10.1016/j.proci.2012.07.043](https://doi.org/10.1016/j.proci.2012.07.043).
- [4] T. Turányi, Sensitivity analysis of complex kinetic systems. Tools and applications, *J. Math. Chem.* 5 (3) (1990) 203–248, doi:[10.1007/BF01166355](https://doi.org/10.1007/BF01166355).
- [5] I.G. Zsély, J. Zádor, T. Turányi, Uncertainty analysis of NO production during methane combustion, *Int. J. Chem. Kinet.* 40 (11) (2008) 754–768, doi:[10.1002/kin.20373](https://doi.org/10.1002/kin.20373).
- [6] T. Ziehn, K.J. Hughes, J.F. Griffiths, R. Porter, A.S. Tomlin, A global sensitivity study of cyclohexane oxidation under low temperature fuel-rich conditions using HDMR methods, *Combust. Theor. Model.* 13 (4) (2009) 589–605, doi:[10.1080/13647830902878398](https://doi.org/10.1080/13647830902878398).
- [7] D.G. Truhlar, B.C. Garrett, S.J. Klippenstein, Current status of transition-state theory, *J. Phys. Chem.* 100 (31) (1996) 12771–12800, doi:[10.1021/jp953748q](https://doi.org/10.1021/jp953748q).
- [8] C.K. Westbrook, F.L. Dryer, Chemical kinetic modeling of hydrocarbon combustion, *Prog. Energy Combust. Sci.* 10 (1) (1984) 1–57, doi:[10.1016/0360-1285\(84\)90118-7](https://doi.org/10.1016/0360-1285(84)90118-7).
- [9] H.J. Curran, P. Gaffuri, W.J. Pitz, C.K. Westbrook, A comprehensive modeling study of n-heptane oxidation, *Combust. Flame* 114 (1) (1998) 149–177, doi:[10.1016/S0010-2180\(97\)00282-4](https://doi.org/10.1016/S0010-2180(97)00282-4).
- [10] H.J. Curran, P. Gaffuri, W.J. Pitz, C.K. Westbrook, A comprehensive modeling study of iso-octane oxidation, *Combust. Flame* 129 (3) (2002) 253–280, doi:[10.1016/S0010-2180\(01\)00373-X](https://doi.org/10.1016/S0010-2180(01)00373-X).
- [11] S.M. Sarathy, S. Vranckx, K. Yasunaga, M. Mehl, P. Oßwald, W.K. Metcalfe, C.K. Westbrook, W.J. Pitz, K. Kohse-Höinghaus, R.X. Fernandes, H.J. Curran, A comprehensive chemical kinetic combustion model for the four butanol isomers, *Combust. Flame* 159 (6) (2012) 2028–2055, doi:[10.1016/j.combustflame.2011.12.017](https://doi.org/10.1016/j.combustflame.2011.12.017).
- [12] E. Ranzi, C. Cavallotti, A. Cuoci, A. Frassoldati, M. Pelucchi, T. Faravelli, New reaction classes in the kinetic modeling of low temperature oxidation of n-alkanes, *Combust. Flame* 162 (5) (2015) 1679–1691, doi:[10.1016/j.combustflame.2014.11.030](https://doi.org/10.1016/j.combustflame.2014.11.030).
- [13] J. Bugler, K.P. Somers, E.J. Silke, J.H. Curran, Revisiting the kinetics and thermodynamics of the low-temperature oxidation pathways of alkanes: a case study of the three pentane isomers, *J. Phys. Chem. A* 119 (28) (2015) 7510–7527, doi:[10.1021/acs.jpca.5b00837](https://doi.org/10.1021/acs.jpca.5b00837).
- [14] K. Zhang, C. Banyon, C. Togbé, P. Dagaut, J. Bugler, H.J. Curran, An experimental and kinetic modeling study of n-hexane oxidation, *Combust. Flame* 162 (11) (2015) 4194–4207, doi:[10.1016/j.combustflame.2015.08.001](https://doi.org/10.1016/j.combustflame.2015.08.001).
- [15] J. Bugler, B. Marks, O. Mathieu, R. Archuleta, A. Camou, C. Grégoire, K.A. Heufer, E.L. Petersen, H.J. Curran, An ignition delay time and chemical kinetic modeling study of the pentane isomers, *Combust. Flame* 163 (2016) 138–156, doi:[10.1016/j.combustflame.2015.09.014](https://doi.org/10.1016/j.combustflame.2015.09.014).
- [16] K. Zhang, C. Banyon, J. Bugler, H.J. Curran, A. Rodriguez, O. Herbinet, F. Battin-Leclerc, C. B'Chir, K.A. Heufer, An updated experimental and kinetic modeling study of n-heptane oxidation, *Combust. Flame* 172 (2016) 116–135, doi:[10.1016/j.combustflame.2016.06.028](https://doi.org/10.1016/j.combustflame.2016.06.028).
- [17] S.Y. Mohamed, L. Cai, F. Khaled, C. Banyon, Z. Wang, M.J. Al Rashidi, H. Pitsch, H.J. Curran, A. Farooq, S.M. Sarathy, Modeling ignition of a heptane isomer: Improved thermodynamics, reaction pathways, kinetics, and rate rule optimizations for 2-methylhexane, *J. Phys. Chem. A* 120 (14) (2016) 2201–2217, doi:[10.1021/acs.jpca.6b00907](https://doi.org/10.1021/acs.jpca.6b00907).
- [18] N. Atef, G. Kukkadapu, S.Y. Mohamed, M. Al Rashidi, C. Banyon, M. Mehl, K.A. Heufer, E.F. Nasir, A. Alfazazi, A.K. Das, C.K. Westbrook, W.J. Pitz, T. Lu, A. Farooq, C.-J. Sung, H.J. Curran, S.M. Sarathy, A comprehensive iso-octane combustion model with improved thermochemistry and chemical kinetics, *Combust. Flame* 178 (2017) 111–134, doi:[10.1016/j.combustflame.2016.12.029](https://doi.org/10.1016/j.combustflame.2016.12.029).
- [19] H.J. Curran, Developing detailed chemical kinetic mechanisms for fuel combustion, *Proc. Combust. Inst.* 37 (1) (2019) 57–81, doi:[10.1016/j.proci.2018.06.054](https://doi.org/10.1016/j.proci.2018.06.054).
- [20] S.W. Benson, J.H. Buss, Additivity rules for the estimation of molecular properties. Thermodynamic properties, *J. Chem. Phys.* 29 (3) (1958) 546–572, doi:[10.1063/1.1744539](https://doi.org/10.1063/1.1744539).
- [21] S.W. Benson, F.R. Cruickshank, D.M. Golden, G.R. Haugen, H.E. O'Neal, A.S. Rodgers, R. Shaw, R. Walsh, Additivity rules for the estimation of thermochemical properties, *Chem. Rev.* 69 (3) (1969) 279–324, doi:[10.1021/cr60259a002](https://doi.org/10.1021/cr60259a002).
- [22] S.W. Benson, *Thermochemical Kinetics: Methods for the Estimation of Thermochemical Data and Rate Parameters*, Wiley-Interscience, Wiley, 1976.
- [23] C.F. Goldsmith, G.R. Magoon, W.H. Green, Database of small molecule thermochemistry for combustion, *J. Phys. Chem. A* 116 (36) (2012) 9033–9057, doi:[10.1021/jp303819e](https://doi.org/10.1021/jp303819e).
- [24] S.J. Klippenstein, L.B. Harding, B. Ruscic, Ab initio computations and Active Thermochemical Tables hand in hand: Heats of formation of core combustion species, *J. Phys. Chem. A* 121 (35) (2017) 6580–6602, doi:[10.1021/acs.jpca.7b05945](https://doi.org/10.1021/acs.jpca.7b05945).
- [25] J.-R. Cao, M.H. Back, The heat of formation of the ethyl radical, *Int. J. Chem. Kinet.* 16 (8) (1984) 961–966, doi:[10.1002/kin.550160803](https://doi.org/10.1002/kin.550160803).

- [26] T. Turányi, L. Zalotai, S. Dóbbé, T. Bérces, Effect of the uncertainty of kinetic and thermodynamic data on methane flame simulation results, *Phys. Chem. Chem. Phys.* 4 (2002) 2568–2578, doi:[10.1039/B109154A](https://doi.org/10.1039/B109154A).
- [27] G.P. Smith, D.M. Golden, M. Frenklach, N.W. Moriarty, B. Eiteneer, M. Goldenberg, C.T. Bowman, K.R. Hanson, S. Song, W.C. Gardiner Jr, V.V. Lissianski, Z. Qin, GRI-Mech 3.0, [http://www.me.berkeley.edu/gri\\_mech/](http://www.me.berkeley.edu/gri_mech/).
- [28] J. Zádor, I.G. Zsély, T. Turányi, M. Ratto, S. Tarantola, A. Saltelli, Local and global uncertainty analyses of a methane flame model, *J. Phys. Chem. A* 109 (43) (2005) 9795–9807, doi:[10.1021/jp053270i](https://doi.org/10.1021/jp053270i).
- [29] K.J. Hughes, J.F. Griffiths, M. Fairweather, A.S. Tomlin, Evaluation of models for the low temperature combustion of alkanes through interpretation of pressure-temperature ignition diagrams, *Phys. Chem. Chem. Phys.* 8 (2006) 3197–3210, doi:[10.1039/B605379C](https://doi.org/10.1039/B605379C).
- [30] S.K. Sirumalla, M.A. Mayer, K.E. Niemeyer, R.H. West, Assessing impacts of discrepancies in model parameters on autoignition model performance: A case study using butanol, *Combust. Flame* 190 (2018) 284–292, doi:[10.1016/j.combustflame.2017.11.018](https://doi.org/10.1016/j.combustflame.2017.11.018).
- [31] F. vom Lehn, L. Cai, H. Pitsch, Sensitivity analysis, uncertainty quantification, and optimization for thermochemical properties in chemical kinetic combustion models, *Proc. Combust. Inst.* 37 (1) (2019a) 771–779, doi:[10.1016/j.proci.2018.06.188](https://doi.org/10.1016/j.proci.2018.06.188).
- [32] F. vom Lehn, L. Cai, H. Pitsch, Impact of thermochemistry on optimized kinetic model predictions: Auto-ignition of diethyl ether, *Combust. Flame* 210 (2019b) 454–466, doi:[10.1016/j.combustflame.2019.09.011](https://doi.org/10.1016/j.combustflame.2019.09.011).
- [33] E.S. Domalski, E.D. Hearing, Estimation of the thermodynamic properties of hydrocarbons at 298.15 K, *J. Phys. Chem. Ref. Data* 17 (4) (1988) 1637–1678, doi:[10.1063/1.555814](https://doi.org/10.1063/1.555814).
- [34] N. Cohen, Revised group additivity values for enthalpies of formation (at 298 K) of carbon-hydrogen and carbon-hydrogen-oxygen compounds, *J. Phys. Chem. Ref. Data* 25 (6) (1996) 1411–1481, doi:[10.1063/1.555988](https://doi.org/10.1063/1.555988).
- [35] I. Marsi, B. Viskolcz, L. Seres, Application of the group additivity method to alkyl radicals: An ab initio study, *J. Phys. Chem. A* 104 (19) (2000) 4497–4504, doi:[10.1021/jp992355e](https://doi.org/10.1021/jp992355e).
- [36] R. Sumathi, W.H. Green, Thermodynamic properties of ketenes: Group additivity values from quantum chemical calculations, *J. Phys. Chem. A* 106 (34) (2002) 7937–7949, doi:[10.1021/jp021179y](https://doi.org/10.1021/jp021179y).
- [37] R. Sumathi, W.H. Green, Oxygenate, oxalkyl and alkoxy carbonyl thermochemistry and rates for hydrogen abstraction from oxygenates, *Phys. Chem. Chem. Phys.* 5 (2003) 3402–3417, doi:[10.1039/B307050F](https://doi.org/10.1039/B307050F).
- [38] C.-C. Chen, J.W. Bozzelli, Structures, intramolecular rotation barriers, and thermochemical properties of methyl ethyl, methyl isopropyl, and methyl tert-butyl ethers and the corresponding radicals, *J. Phys. Chem. A* 107 (22) (2003) 4531–4546, doi:[10.1021/jp022131n](https://doi.org/10.1021/jp022131n).
- [39] N. Sebban, J.W. Bozzelli, H. Bockhorn, Thermochemical properties, rotation barriers, bond energies, and group additivity for vinyl, phenyl, ethynyl, and allyl peroxides, *J. Phys. Chem. A* 108 (40) (2004) 8353–8366, doi:[10.1021/jp031067m](https://doi.org/10.1021/jp031067m).
- [40] M.K. Sabbe, M. Saeys, M.-F. Reyniers, G.B. Marin, V. van Speybroeck, M. Waroquier, Group additive values for the gas phase standard enthalpy of formation of hydrocarbons and hydrocarbon radicals, *J. Phys. Chem. A* 109 (33) (2005) 7466–7480, doi:[10.1021/jp050484r](https://doi.org/10.1021/jp050484r).
- [41] M.K. Sabbe, F. De Vleeschouwer, M.-F. Reyniers, M. Waroquier, G.B. Marin, First principles based group additive values for the gas phase standard entropy and heat capacity of hydrocarbons and hydrocarbon radicals, *J. Phys. Chem. A* 112 (47) (2008) 12235–12251, doi:[10.1021/jp807526n](https://doi.org/10.1021/jp807526n).
- [42] P.D. Paraskevas, M.K. Sabbe, M. Reyniers, N. Papayannakos, B. Marin G., Group additive values for the gas phase standard enthalpy of formation, entropy and heat capacity of oxygenates, *Chem. Eur. J.* 19 (2013) 16431–16452, doi:[10.1002/chem.201301381](https://doi.org/10.1002/chem.201301381).
- [43] S.M. Burke, J.M. Simmie, H.J. Curran, Critical evaluation of thermochemical properties of C1–C4 species: updated group-contributions to estimate thermochemical properties, *J. Phys. Chem. Ref. Data* 44 (1) (2015) 013101, doi:[10.1063/1.4902535](https://doi.org/10.1063/1.4902535).
- [44] A. Ince, H. Carstensen, M. Reyniers, G.B. Marin, First-principles based group additivity values for thermochemical properties of substituted aromatic compounds, *AIChE J.* 61 (2015) 3858–3870.
- [45] Y. Li, H.J. Curran, Extensive theoretical study of the thermochemical properties of unsaturated hydrocarbons and allylic and super-allylic radicals: The development and optimization of group additivity values, *J. Phys. Chem. A* 122 (20) (2018) 4736–4749, doi:[10.1021/acs.jpca.8b02912](https://doi.org/10.1021/acs.jpca.8b02912).
- [46] H. Janbazi, O. Hasemann, C. Schulz, A. Kempf, I. Wlokas, S. Peukert, Response surface and group additivity methodology for estimation of thermodynamic properties of organosilanes, *Int. J. Chem. Kinet.* 50 (9) (2018) 681–690, doi:[10.1002/kin.21192](https://doi.org/10.1002/kin.21192).
- [47] C.W. Gao, Automatic reaction mechanism generation: High fidelity predictive modeling of combustion processes, Ph.D. thesis, Massachusetts Institute of Technology, 2016. <https://dspace.mit.edu/handle/1721.1/104205>
- [48] A. Fridlyand, M.S. Johnson, S.S. Goldsborough, R.H. West, M.J. McNenly, M. Mehl, W.J. Pitz, The role of correlations in uncertainty quantification of transportation relevant fuel models, *Combust. Flame* 180 (2017) 239–249, doi:[10.1016/j.combustflame.2016.10.014](https://doi.org/10.1016/j.combustflame.2016.10.014).
- [49] M. Hantouche, S.M. Sarathy, O.M. Knio, Global sensitivity analysis of n-butanol reaction kinetics using rate rules, *Combust. Flame* 196 (2018) 452–465, doi:[10.1016/j.combustflame.2018.06.035](https://doi.org/10.1016/j.combustflame.2018.06.035).
- [50] M. Frenklach, Systematic optimization of a detailed kinetic model using a methane ignition example, *Combust. Flame* 58 (1) (1984) 69–72, doi:[10.1016/0010-2180\(84\)90079-8](https://doi.org/10.1016/0010-2180(84)90079-8).
- [51] D.A. Sheen, H. Wang, The method of uncertainty quantification and minimization using polynomial chaos expansions, *Combust. Flame* 158 (12) (2011) 2358–2374, doi:[10.1016/j.combustflame.2011.05.010](https://doi.org/10.1016/j.combustflame.2011.05.010).
- [52] L. Cai, H. Pitsch, Mechanism optimization based on reaction rate rules, *Combust. Flame* 161 (2) (2014) 405–415, doi:[10.1016/j.combustflame.2013.08.024](https://doi.org/10.1016/j.combustflame.2013.08.024).
- [53] S.M. Sarathy, C.K. Westbrook, M. Mehl, W.J. Pitz, C. Togbe, P. Dagaut, H. Wang, M.A. Oehlschlaeger, U. Niemann, K. Seshadri, S. Veloo P., C. Ji, F.N. Egolfopoulos, T. Lu, Comprehensive chemical kinetic modeling of the oxidation of 2-methylalkanes from C7 to C20, *Combust. Flame* 158 (12) (2011) 2338–2357, doi:[10.1016/j.combustflame.2011.05.007](https://doi.org/10.1016/j.combustflame.2011.05.007).
- [54] H. Pitsch, FlameMaster: A C++ computer program for 0D combustion and 1D laminar flame calculations, <https://www.itv.rwth-aachen.de/index.php?id=flamemaster>.
- [55] R. Sivaramakrishnan, J.V. Michael, Rate constants for OH with selected large alkanes: shock-tube measurements and an improved group scheme, *J. Phys. Chem. A* 113 (17) (2009) 5047–5060, doi:[10.1021/jp810987u](https://doi.org/10.1021/jp810987u).
- [56] J. Badra, A. Elwardany, A. Farooq, Shock tube measurements of the rate constants for seven large alkanes + OH, *Proc. Combust. Inst.* 35 (1) (2015) 189–196, doi:[10.1016/j.proci.2014.05.098](https://doi.org/10.1016/j.proci.2014.05.098).
- [57] L. Cai, H. Pitsch, Optimized chemical mechanism for combustion of gasoline surrogate fuels, *Combust. Flame* 162 (5) (2015) 1623–1637, doi:[10.1016/j.combustflame.2014.11.018](https://doi.org/10.1016/j.combustflame.2014.11.018).
- [58] L. Cai, H. Pitsch, S.Y. Mohamed, V. Raman, J. Bugler, H.J. Curran, S.M. Sarathy, Optimized reaction mechanism rate rules for ignition of normal alkanes, *Combust. Flame* 173 (2016) 468–482, doi:[10.1016/j.combustflame.2016.04.022](https://doi.org/10.1016/j.combustflame.2016.04.022).
- [59] Y. Chang, M. Jia, B. Niu, Y. Zhang, M. Xie, Y. Li, Construction and assessment of reduced oxidation mechanisms using global sensitivity analysis and uncertainty analysis, *Proc. Combust. Inst.* 37 (1) (2019) 751–761, doi:[10.1016/j.proci.2018.07.006](https://doi.org/10.1016/j.proci.2018.07.006).
- [60] É. Hébrard, A.S. Tomlin, R. Bounaceur, F. Battin-Leclerc, Determining predictive uncertainties and global sensitivities for large parameter systems: a case study for n-butane oxidation, *Proc. Combust. Inst.* 35 (1) (2015) 607–616, doi:[10.1016/j.proci.2014.06.027](https://doi.org/10.1016/j.proci.2014.06.027).
- [61] S.P. Verevkin, V.N. Emel'yanenko, V. Diky, C.D. Muzny, R.D. Chirico, M. Frenkel, New group-contribution approach to thermochemical properties of organic compounds: Hydrocarbons and oxygen-containing compounds, *J. Phys. Chem. Ref. Data* 42 (3) (2013) 033102, doi:[10.1063/1.4815957](https://doi.org/10.1063/1.4815957).
- [62] T. He, S. Li, Y. Chi, H.-B. Zhang, Z. Wang, B. Yang, X. He, X. You, An adaptive distance-based group contribution method for thermodynamic property prediction, *Phys. Chem. Chem. Phys.* 18 (2016) 23822–23830, doi:[10.1039/C6CP02929A](https://doi.org/10.1039/C6CP02929A).
- [63] K. Han, A. Jamal, C.A. Grambow, Z.J. Buras, W.H. Green, An extended group additivity method for polycyclic thermochemistry estimation, *Int. J. Chem. Kinet.* 50 (4) (2018) 294–303, doi:[10.1002/kin.21158](https://doi.org/10.1002/kin.21158).
- [64] J. Wu, H. Ning, L. Ma, P. Zhang, W. Ren, Cascaded group-additivity ONIOM: a new method to approach CCSD(T)/CBS energies of large aliphatic hydrocarbons, *Combust. Flame* 201 (2019) 31–43, doi:[10.1016/j.combustflame.2018.12.012](https://doi.org/10.1016/j.combustflame.2018.12.012).
- [65] N. Cohen, S.W. Benson, The thermochemistry of alkanes and cycloalkanes, in: S. Patai, Z. Rappoport (Eds.), *Alkanes and Cycloalkanes*, John Wiley & Sons, Ltd (1992), pp. 215–287, doi:[10.1002/0470034378.ch6](https://doi.org/10.1002/0470034378.ch6).
- [66] E.R. Ritter, J.W. Bozzelli, THERM: Thermodynamic property estimation for gas phase radicals and molecules, *Int. J. Chem. Kinet.* 23 (9) (1991) 767–778, doi:[10.1002/kin.550230903](https://doi.org/10.1002/kin.550230903).
- [67] E.R. Ritter, THERM: A computer code for estimating thermodynamic properties for species important to combustion and reaction modeling, *J. Chem. Inf. Comput. Sci.* 31 (1991) 400–408, doi:[10.1021/ci00003a006](https://doi.org/10.1021/ci00003a006).
- [68] C. Muller, V. Michel, G. Scacchi, G.M. Côme, THERGAS: a computer program for the evaluation of thermochemical data of molecules and free radicals in the gas phase, *J. Chim. Phys.* 92 (1995) 1154–1178, doi:[10.1051/jcp/1995921154](https://doi.org/10.1051/jcp/1995921154).
- [69] C.W. Gao, J.W. Allen, W.H. Green, R.H. West, Reaction mechanism generator: automatic construction of chemical kinetic mechanisms, *Comput. Phys. Commun.* 203 (2016) 212–225, doi:[10.1016/j.cpc.2016.02.013](https://doi.org/10.1016/j.cpc.2016.02.013).
- [70] S.M. Villano, L.K. Huynh, H.-H. Carstensen, A.M. Dean, High-pressure rate rules for alkyl + O<sub>2</sub> reactions. 1. The dissociation, concerted elimination, and isomerization channels of the alkyl peroxy radical, *J. Phys. Chem. A* 115 (46) (2011) 13425–13442, doi:[10.1021/jp2079204](https://doi.org/10.1021/jp2079204).
- [71] T.H. Lay, J.W. Bozzelli, A.M. Dean, E.R. Ritter, Hydrogen atom bond increments for calculation of thermodynamic properties of hydrocarbon radical species, *J. Phys. Chem.* 99 (39) (1995) 14514–14527, doi:[10.1021/j100039a045](https://doi.org/10.1021/j100039a045).
- [72] A. Burcat, Thermochemical data for combustion calculations, in: W.C. Gardiner (Ed.), *Combustion Chemistry*, Springer, New York (1984), pp. 455–473, doi:[10.1007/978-1-4684-0186-8\\_8](https://doi.org/10.1007/978-1-4684-0186-8_8).
- [73] B. Ruscic, R.E. Pinzon, M.L. Morton, G. von Laszewski, S.J. Bittner, S.G. Nijssure, K.A. Amin, M. Minkoff, A.F. Wagner, Introduction to Active Thermochemical Tables: Several key enthalpies of formation revisited, *J. Phys. Chem. A* 108 (45) (2004) 9979–9997, doi:[10.1021/jp047912y](https://doi.org/10.1021/jp047912y).
- [74] A.G. Császár, T. Furtenbacher, From a network of computed reaction enthalpies to atom-based thermochemistry (NEAT), *Chem. Eur. J.* 16 (16) (2010) 4826–4835, doi:[10.1002/chem.200903252](https://doi.org/10.1002/chem.200903252).
- [75] M.P. Burke, S.J. Klippenstein, L.B. Harding, A quantitative explanation for the apparent anomalous temperature dependence of OH+HO<sub>2</sub>=H<sub>2</sub>O+O<sub>2</sub> through

- multi-scale modeling, *Proc. Combust. Inst.* 34 (1) (2013) 547–555, doi:[10.1016/j.proci.2012.05.041](https://doi.org/10.1016/j.proci.2012.05.041).
- [76] M.P. Burke, C.F. Goldsmith, S.J. Klippenstein, O. Welz, H. Huang, I.O. Antonov, J.D. Savee, D.L. Osborn, J. Zádor, C.A. Taatjes, L. Sheps, Multiscale informatics for low-temperature propane oxidation: Further complexities in studies of complex reactions, *J. Phys. Chem. A* 119 (28) (2015) 7095–7115, doi:[10.1021/acs.jpca.5b01003](https://doi.org/10.1021/acs.jpca.5b01003).
- [77] B. Ruscic, Uncertainty quantification in thermochemistry, benchmarking electronic structure computations, and Active Thermochemical Tables, *Int. J. Quantum Chem.* 114 (17) (2014) 1097–1101, doi:[10.1002/qua.24605](https://doi.org/10.1002/qua.24605).
- [78] M. Frenklach, Transforming data into knowledge—Process Informatics for combustion chemistry, *Proc. Combust. Inst.* 31 (1) (2007) 125–140, doi:[10.1016/j.proci.2006.08.121](https://doi.org/10.1016/j.proci.2006.08.121).
- [79] Y. Tao, G.P. Smith, H. Wang, Critical kinetic uncertainties in modeling hydrogen/carbon monoxide, methane, methanol, formaldehyde, and ethylene combustion, *Combust. Flame* 195 (2018) 18–29, doi:[10.1016/j.combustflame.2018.02.006](https://doi.org/10.1016/j.combustflame.2018.02.006).
- [80] M. Frenklach, H. Wang, M.J. Rabinowitz, Optimization and analysis of large chemical kinetic mechanisms using the solution mapping method—combustion of methane, *Prog. Energy Combust. Sci.* 18 (1) (1992) 47–73, doi:[10.1016/0360-1285\(92\)90032-V](https://doi.org/10.1016/0360-1285(92)90032-V).
- [81] Y. Tao, H. Wang, Joint probability distribution of Arrhenius parameters in reaction model optimization and uncertainty minimization, *Proc. Combust. Inst.* 37 (1) (2019) 817–824, doi:[10.1016/j.proci.2018.08.052](https://doi.org/10.1016/j.proci.2018.08.052).
- [82] T. Turányi, T. Nagy, I.G. Zsély, M. Cserháti, T. Varga, B.T. Szabó, I. Sedyó, T. Kiss P., A. Zempléni, H.J. Curran, Determination of rate parameters based on both direct and indirect measurements, *Int. J. Chem. Kinet.* 44 (2012) 284–302, doi:[10.1002/kin.20717](https://doi.org/10.1002/kin.20717).
- [83] T. Nagy, E. Valkó, I. Sedyó, I.G. Zsély, M.J. Pilling, T. Turányi, Uncertainty of the rate parameters of several important elementary reactions of the H<sub>2</sub> and syngas combustion systems, *Combust. Flame* 162 (5) (2015) 2059–2076, doi:[10.1016/j.combustflame.2015.01.005](https://doi.org/10.1016/j.combustflame.2015.01.005).
- [84] C. Olm, T. Varga, E. Valkó, H.J. Curran, T. Turányi, Uncertainty quantification of a newly optimized methanol and formaldehyde combustion mechanism, *Combust. Flame* 186 (2017) 45–64, doi:[10.1016/j.combustflame.2017.07.029](https://doi.org/10.1016/j.combustflame.2017.07.029).
- [85] K. Braman, T.A. Oliver, V. Raman, Bayesian analysis of syngas chemistry models, *Combust. Theor. Model.* 17 (5) (2013) 858–887, doi:[10.1080/13647830.2013.811541](https://doi.org/10.1080/13647830.2013.811541).
- [86] J. Prager, H.N. Najm, K. Sargsyan, C. Safta, W.J. Pitz, Uncertainty quantification of reaction mechanisms accounting for correlations introduced by rate rules and fitted Arrhenius parameters, *Combust. Flame* 160 (9) (2013) 1583–1593, doi:[10.1016/j.combustflame.2013.01.008](https://doi.org/10.1016/j.combustflame.2013.01.008).
- [87] M. Fürst, P. Sabia, M.L. Lavadera, G. Aversano, M. de Joannon, A. Frassoldati, A. Parente, Optimization of chemical kinetics for methane and biomass pyrolysis products in moderate or intense low-oxygen dilution combustion, *Energy Fuels* 32 (10) (2018) 10194–10201, doi:[10.1021/acs.energyfuels.8b01022](https://doi.org/10.1021/acs.energyfuels.8b01022).
- [88] S.G. Davis, A.B. Mhadeshwar, D.G. Vlachos, H. Wang, A new approach to response surface development for detailed gas-phase and surface reaction kinetic model optimization, *Int. J. Chem. Kinet.* 36 (2) (2004) 94–106, doi:[10.1002/kin.10177](https://doi.org/10.1002/kin.10177).
- [89] O. Park, P.S. Veloo, D.A. Sheen, Y. Tao, F.N. Egolfopoulos, H. Wang, Chemical kinetic model uncertainty minimization through laminar flame speed measurements, *Combust. Flame* 172 (2016) 136–152, doi:[10.1016/j.combustflame.2016.07.004](https://doi.org/10.1016/j.combustflame.2016.07.004).